

Augmented Reality

Introduction to AR Foundation and AR Core

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**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Basic Setup for AR Foundation (+AR Core)

Basic Setup for AR Foundation (+AR Core)

1. Create a New Project
2. Install AR Foundation from the Package Manager
3. Install AR Core from the Package Manager
4. Enable XR Simulator

- Search Core templates
-  Universal 2D
Core

 Universal 3D
Core

 High Definition 3D
Core

 Small Scale Competitive Multiplayer
Core

 3D Mobile
Core

 2D Mobile
Core

 Mixed Reality (MR)
Core

 VR
Core

 AR Mobile
Core

 Competitive Action Multiplayer
Core

 Universal 3D
This template includes the settings and assets you need to start creating with the Universal Render Pipeline (URP).

 Read more

PROJECT SETTINGS

Project name
Lecture_ARFoundation

Location
C:\Users\ManuelaChessa\OneDrive - ...

Unity Organization
manuelachessa

☒ Connect to Unity Cloud ?

☐ Use Unity Version Control ?

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Console

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Unit

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Panels >

Next Window Ctrl+Tab

Previous Window Ctrl+Shift+Tab

Layouts >

Multiplayer >

Unity Version Control

Search >

Asset Store

My Assets

Package Manager

Text >

TextMeshPro >

General >

Rendering >

Animation >

Audio >

Sequencing >

Analysis >

Asset Management >

Accessibility >

2D >

AI >

UI Toolkit >

Visual Scripting >

Assets



Scenes



Settings



TutorialInfo

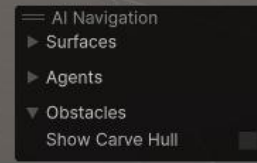
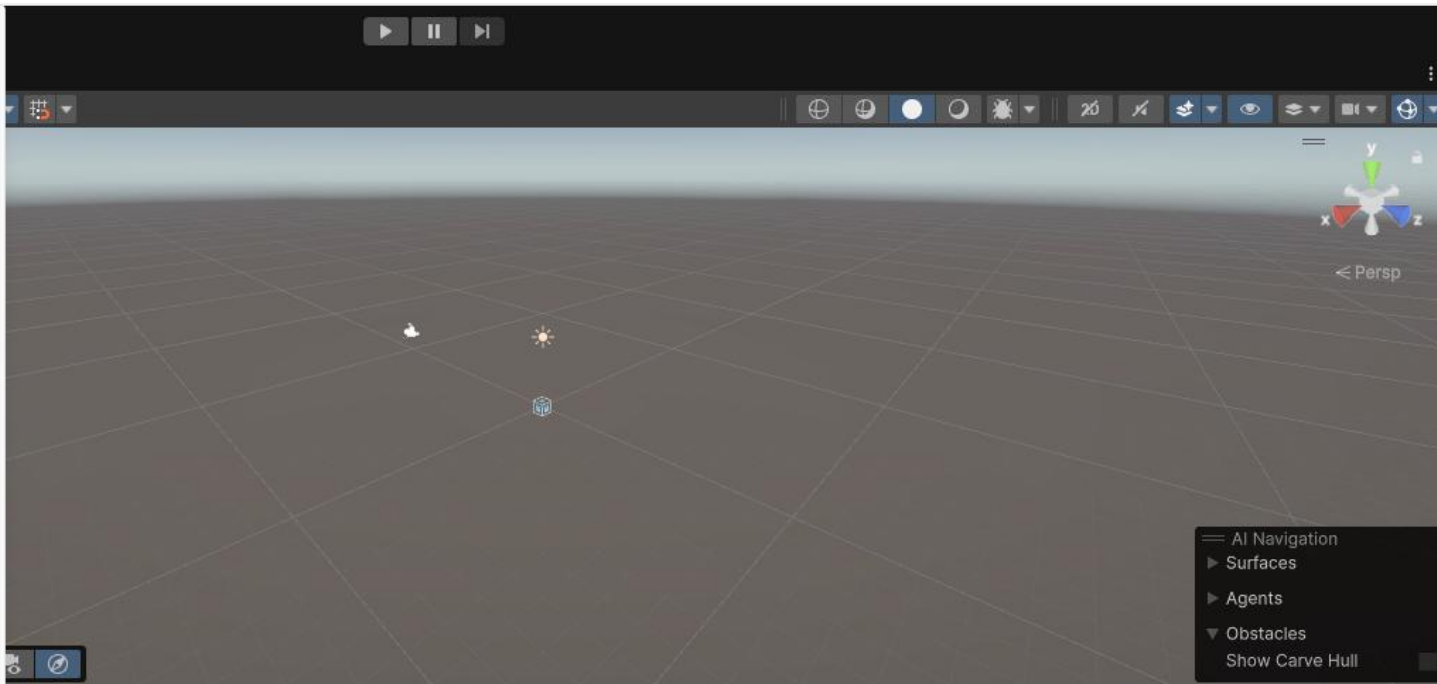


InputSyste...



Readme

Assets/Readme.asset



Inspector



URP
Empty
Template

Welcome to the Universal Render Pipeline

This template includes the settings and assets you need to start creating with the Universal Render Pipeline.

URP Documentation

[Read more about URP](#)

Forums

[Get answers and support](#)

Report bugs

[Submit a report](#)

Remove Readme Assets

Asset Labels

AssetBundle

None

Nor

Package Manager

Package Manager

+ ▾

Sort: Name (asc) ▾

Filters ▾

Clear Filters

🔍 AR 1

✕

▼ Packages

AR Foundation6.0.5

AR Foundation

6.0.5 · January 25, 2025

From **Unity Registry** by Unity Technologies Inc.

com.unity.xr.arfoundation

[Documentation](#) [Changelog](#) [Licenses](#)

Install

Description

Version History

Dependencies

AR Foundation enables you to create multi-platform augmented reality (AR) apps with Unity. To learn more about AR Foundation, click the Documentation link above.

Last refresh May 2, 11:24

↻

Package Manager

Package Manager

+

Sort: Name (asc)

Filters

Clear Filters

🔍 ARCore

▼ Packages

Google ARCore XR Plugin

6.0.5

Google ARCore XR Plugin

6.0.5 · January 25, 2025

From **Unity Registry** by Unity Technologies Inc.

com.unity.xr.arcore

[Documentation](#) | [Changelog](#) | [Licenses](#)

Install

Description

Version History

Dependencies

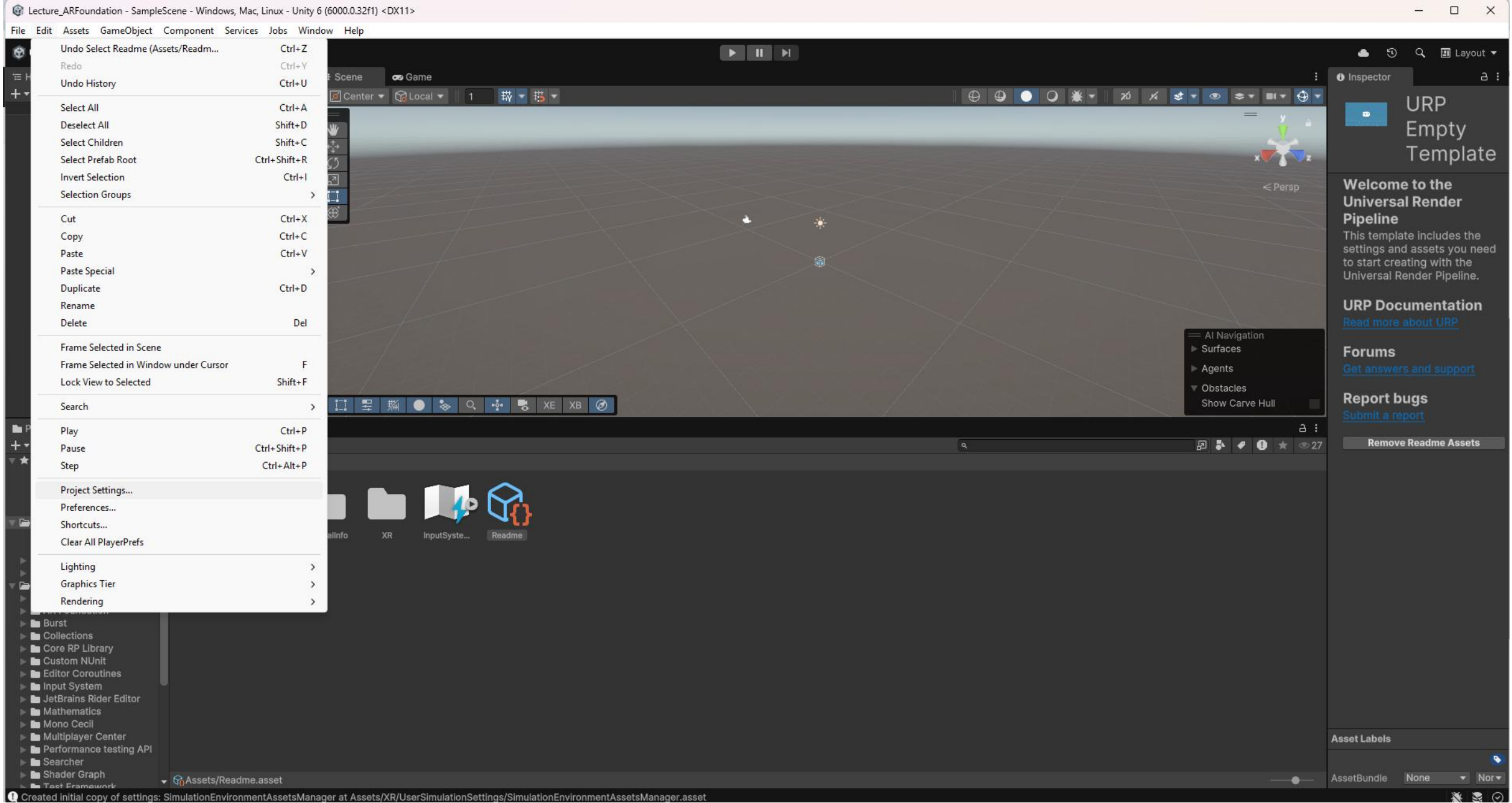
Provides native Google ARCore integration for use with Unity's multi-platform XR API.

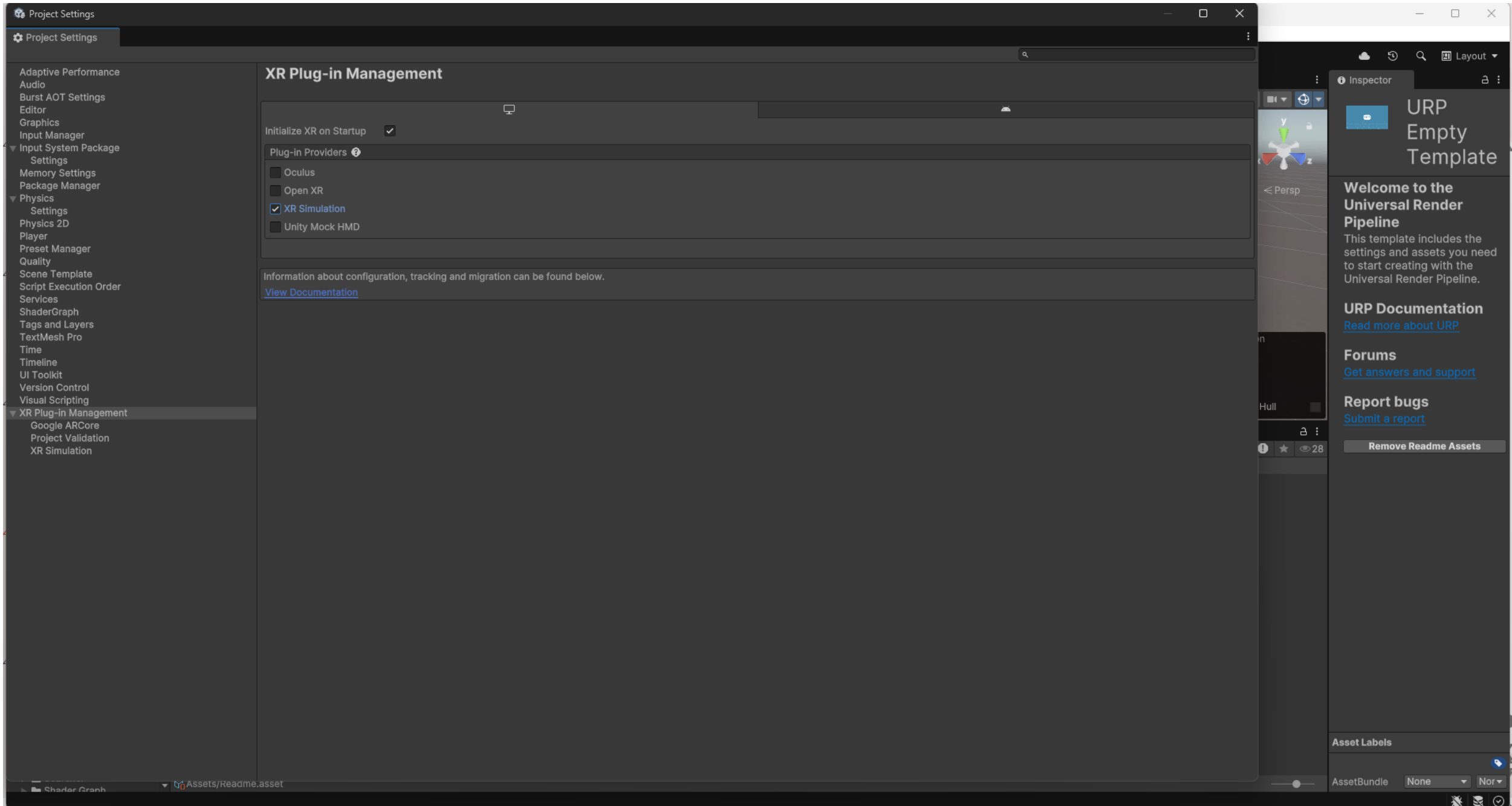
Supports the following features:

- Efficient Background Rendering
- Horizontal Planes
- Depth Data
- Anchors
- Hit Testing
- Occlusion

Last refresh May 2, 11:24

↻

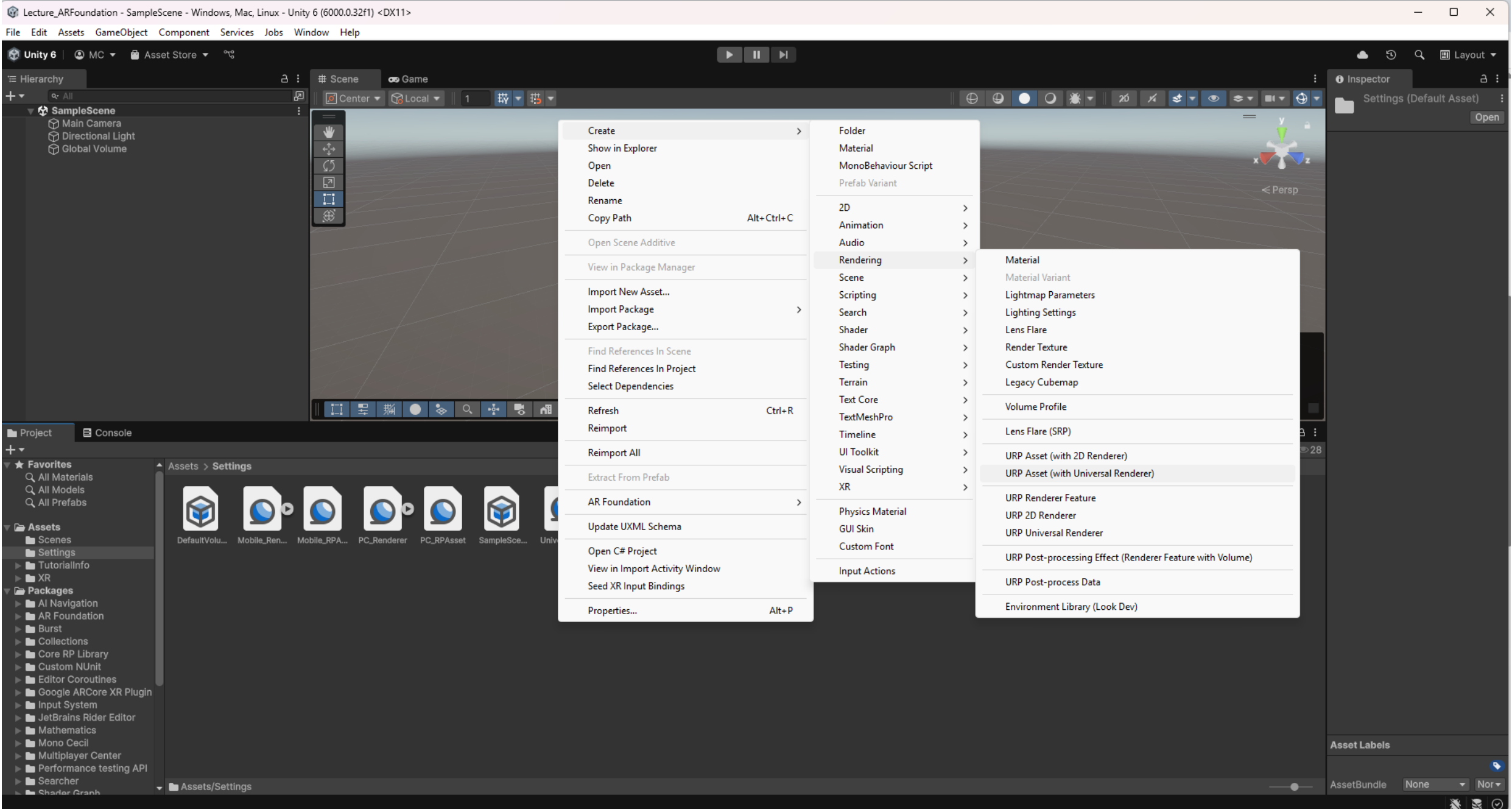


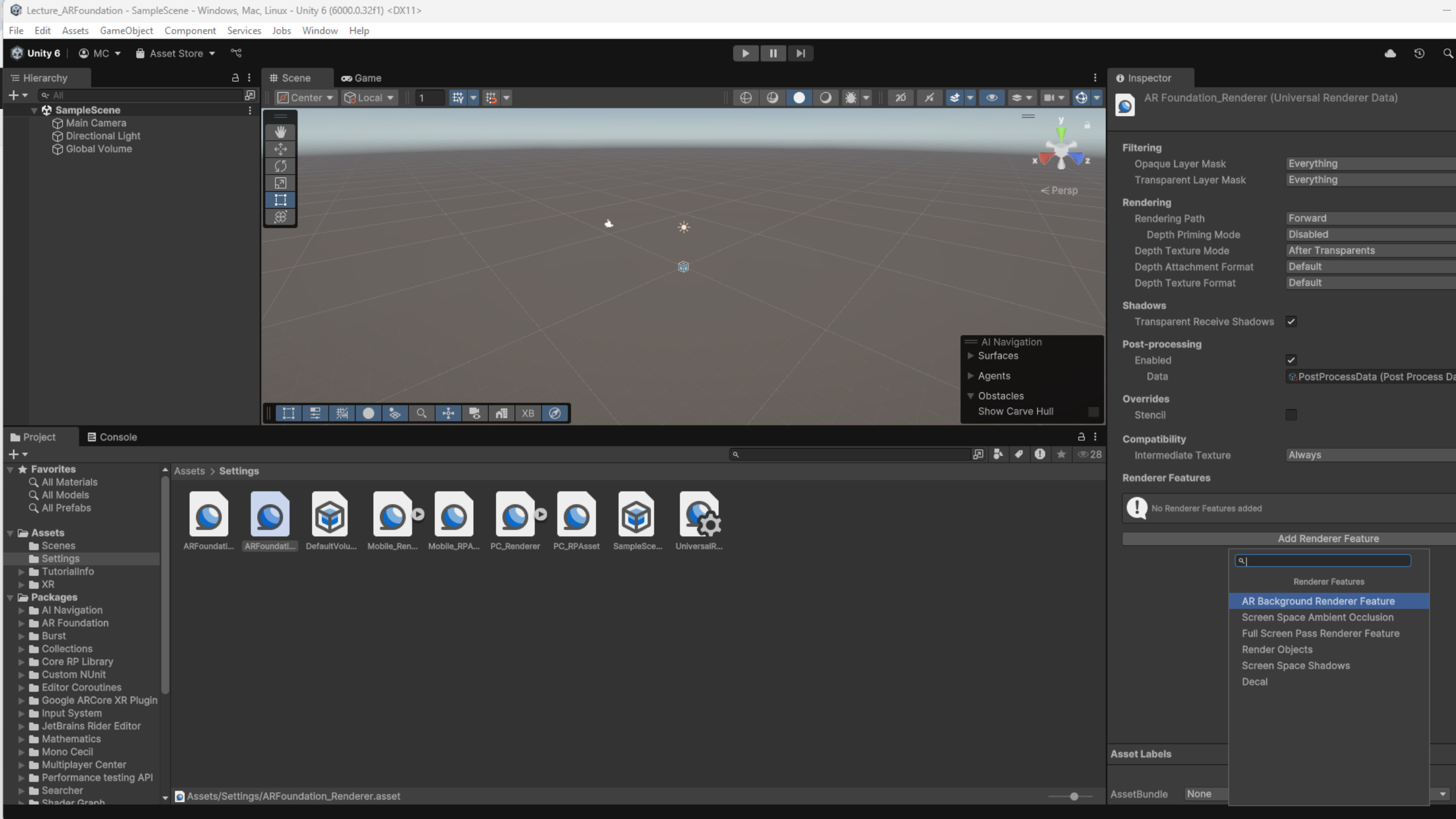


Basic Setup for AR Foundation (+AR Core)

Change the Renderer

<https://docs.unity3d.com/Packages/com.unity.xr.foundation@6.1/manual/project-setup/universal-render-pipeline.html>





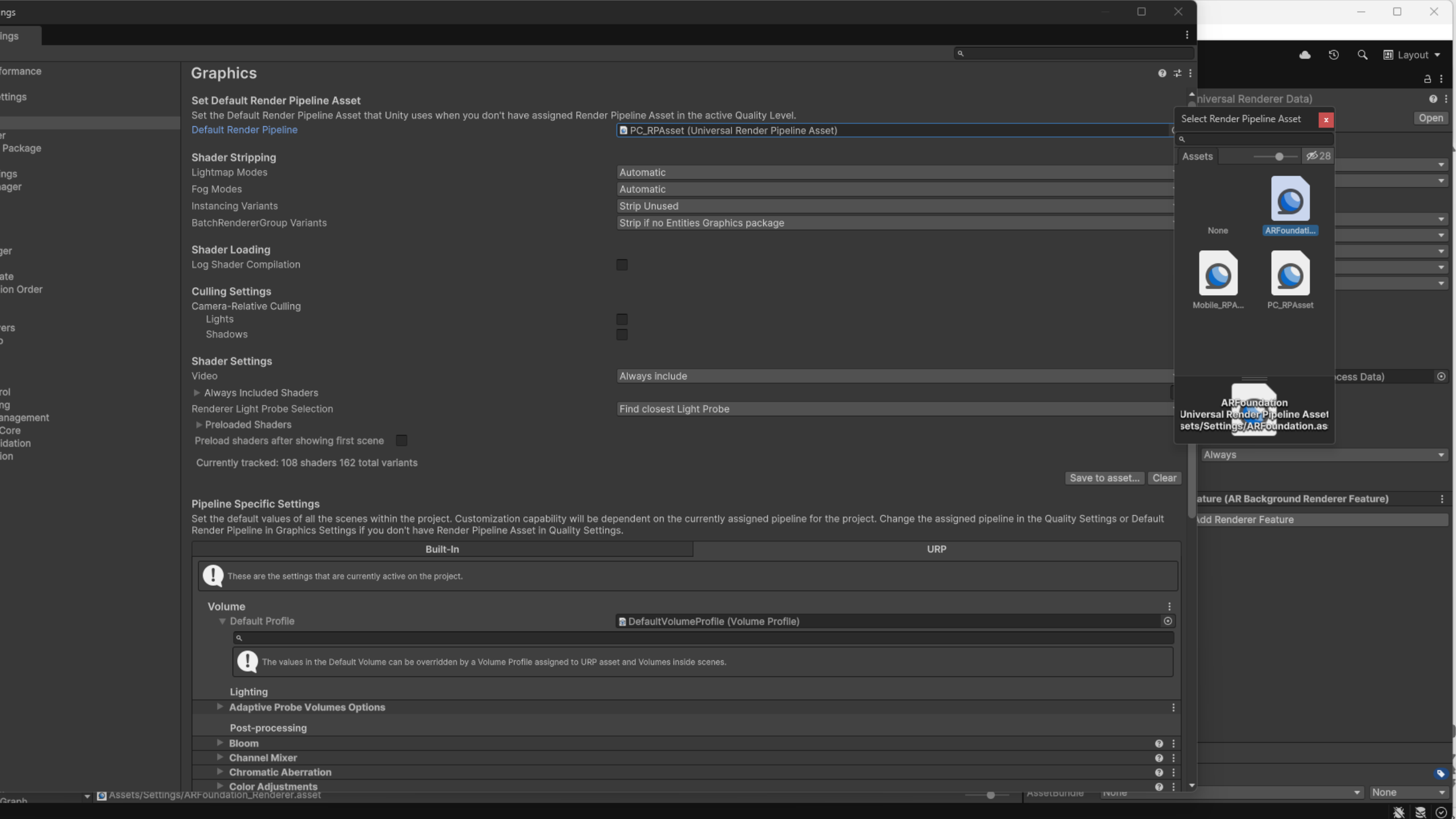
The screenshot displays the Unity 6 development environment. The top menu bar includes File, Edit, Assets, GameObject, Component, Services, Jobs, Window, and Help. The main workspace is divided into several panels:

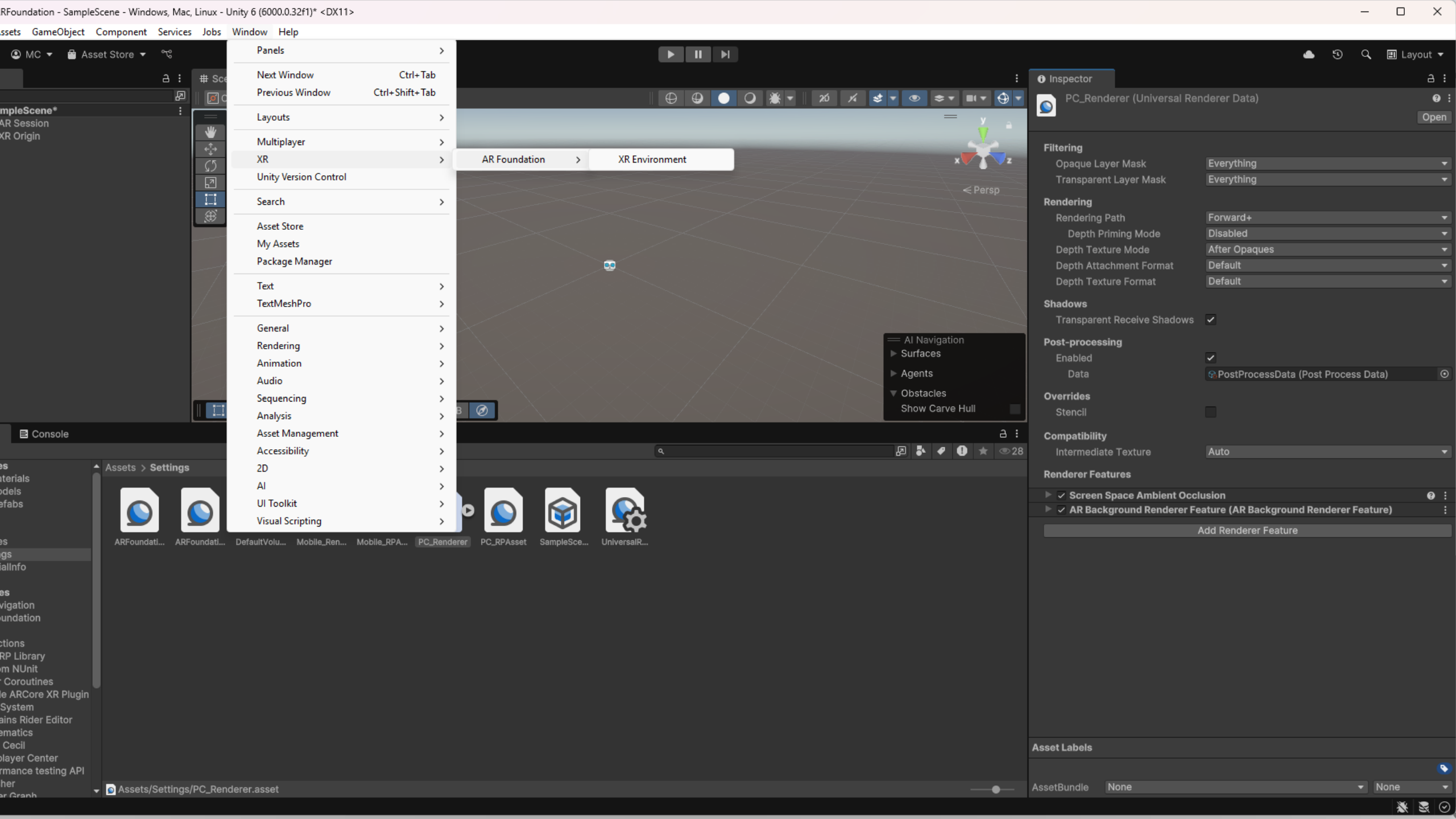
- Hierarchy Panel:** Shows the scene hierarchy with 'SampleScene*' as the root, containing 'AR Session' and 'XR Origin'.
- Inspector Panel:** Displays the 'PC_Renderer (Universal Renderer Data)' settings. The 'Rendering' section is expanded, showing options like 'Rendering Path' (Forward+), 'Depth Priming Mode' (Disabled), 'Depth Texture Mode' (After Opaques), 'Depth Attachment Format' (Default), and 'Depth Texture Format' (Default). The 'Shadows' section shows 'Transparent Receive Shadows' checked. The 'Post-processing' section shows 'Enabled' checked and 'Data' set to 'PostProcessData (Post Process Data)'. The 'Compatibility' section shows 'Intermediate Texture' set to 'Auto'. The 'Renderer Features' section shows 'Screen Space Ambient Occlusion' and 'AR Background Renderer Feature (AR Background Renderer Feature)' checked.
- Project Panel:** Shows the 'Assets' folder structure, including 'Scenes', 'Settings', 'TutorialInfo', 'XR', 'Packages', and 'Assets'. The 'Assets' folder is expanded, showing various assets like 'ARFoundati...', 'DefaultVolu...', 'Mobile_Ren...', 'Mobile_RPA...', 'PC_Renderer', 'PC_RPAsset', 'SampleSce...', and 'UniversalR...'. The 'PC_Renderer' asset is selected.

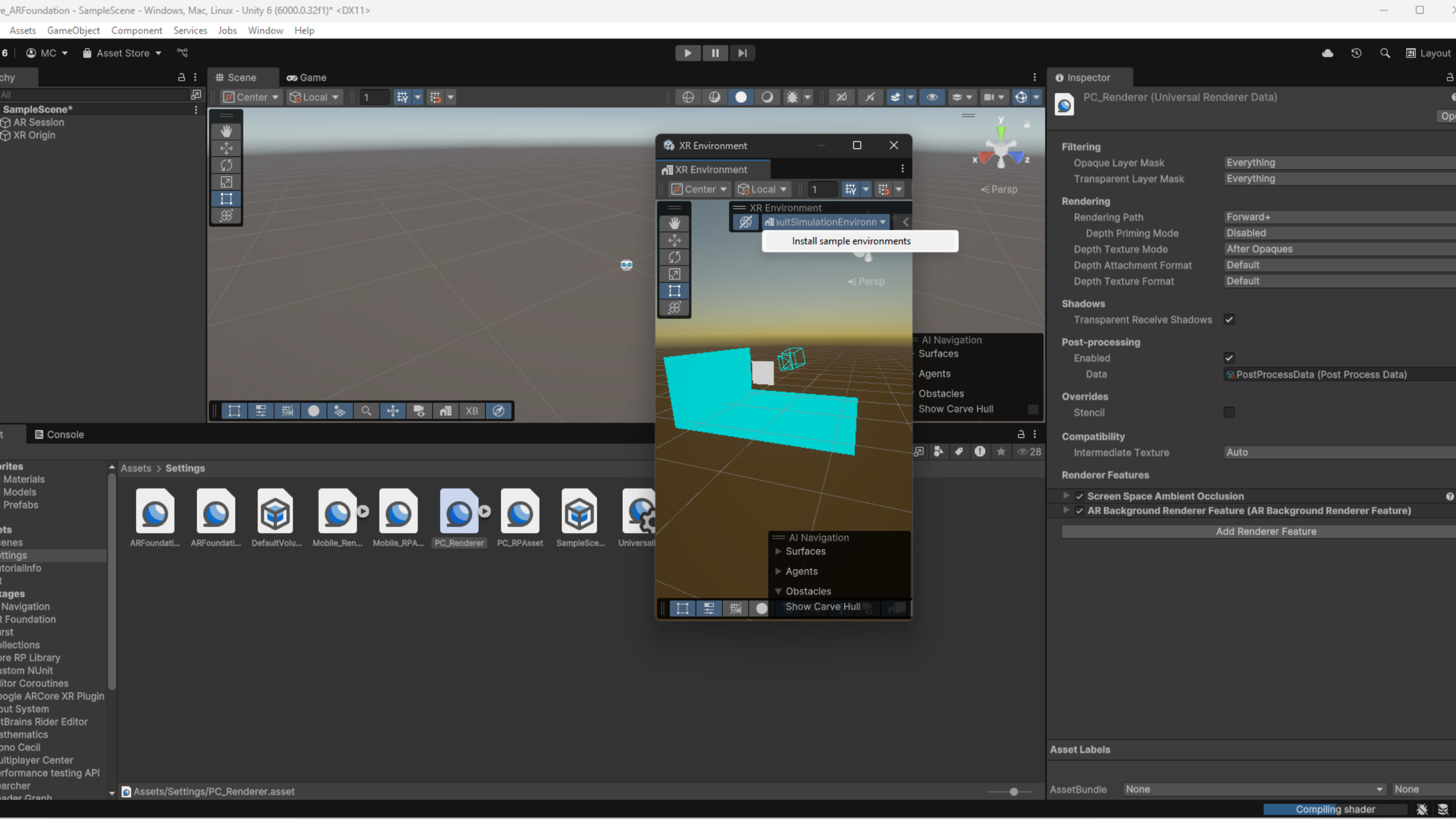
The main viewport shows a 3D scene with a grid floor and a camera view. The camera is positioned at the center of the grid, looking towards the horizon. The scene is rendered in a perspective view.

Basic Setup for AR Foundation (+AR Core)

Setup the Simulator Environments







Unity 6 | MC | Asset Store

Hierarchy | # Scene | Game | Display 1 | Free Aspect | Scale 1.5x | Play Focused | Stats | Gizmos

Inspector | PC_Renderer (Universal Renderer Data)

Filtering

- Opaque Layer Mask: Everything
- Transparent Layer Mask: Everything

Rendering

- Rendering Path: Forward+
- Depth Priming Mode: Disabled
- Depth Texture Mode: After Opaques
- Depth Attachment Format: Default
- Depth Texture Format: Default

Shadows

- Transparent Receive Shadows: ☒

Post-processing

- Enabled: ☒
- Data: PostProcessData (Post Process Data)

Overrides

- Stencil: ☐

Compatibility

- Intermediate Texture: Auto

Renderer Features

- ☒ Screen Space Ambient Occlusion
- ☒ AR Background Renderer Feature (AR Background Renderer Feature)

Add Renderer Feature

Asset Labels

- AssetBundle: None

Project | Console

Assets > Settings

ARFoundati... ARFoundati... DefaultVolu... Mobile_Ren... Mobile_RPA... PC_Renderer PC_RPAsset SampleSce... UniversalR...

Assets

- Scenes
- Settings
- TutorialInfo
- UnityXRContent
- XR

Packages

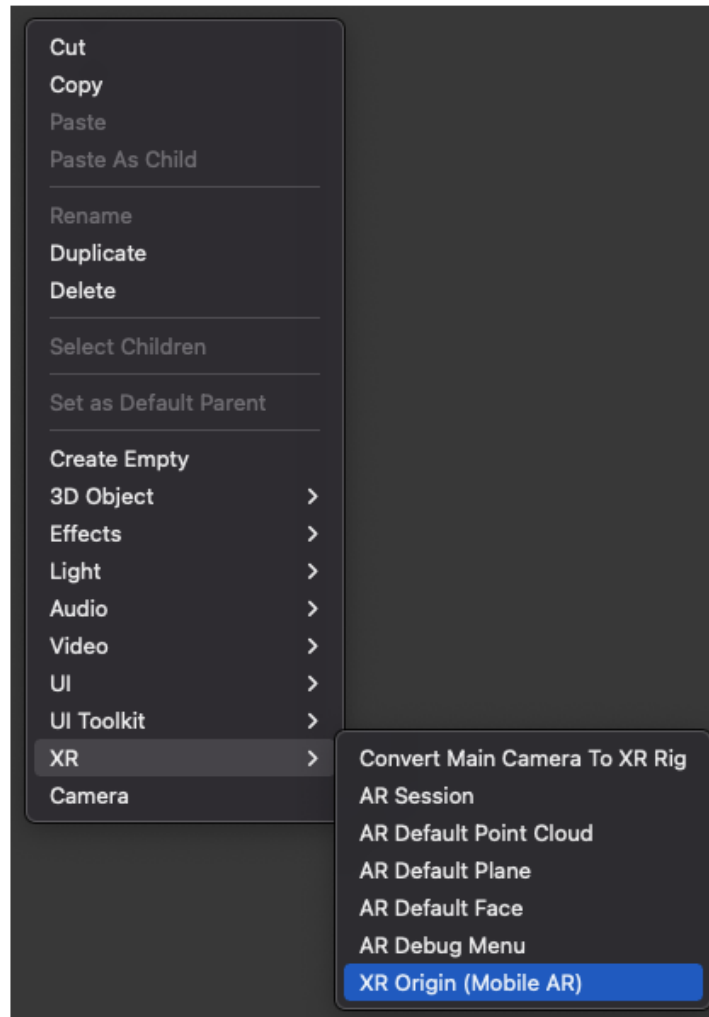
- AI Navigation
- AR Foundation
- Burst
- Collections
- Core RP Library
- Custom NUnit
- Editor Coroutines
- Google ARCore XR Plugin
- Input System
- JetBrains Rider Editor
- Mathematics
- Mono Cecil
- Multiplayer Center
- Performance testing API
- Searcher

Assets/Settings/PC_Renderer.asset

Basic Setup for AR Foundation (+AR Core)

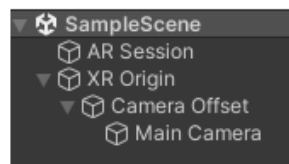
Scene Setup

- XR > AR Session
- XR > XR Origin (Mobile AR)



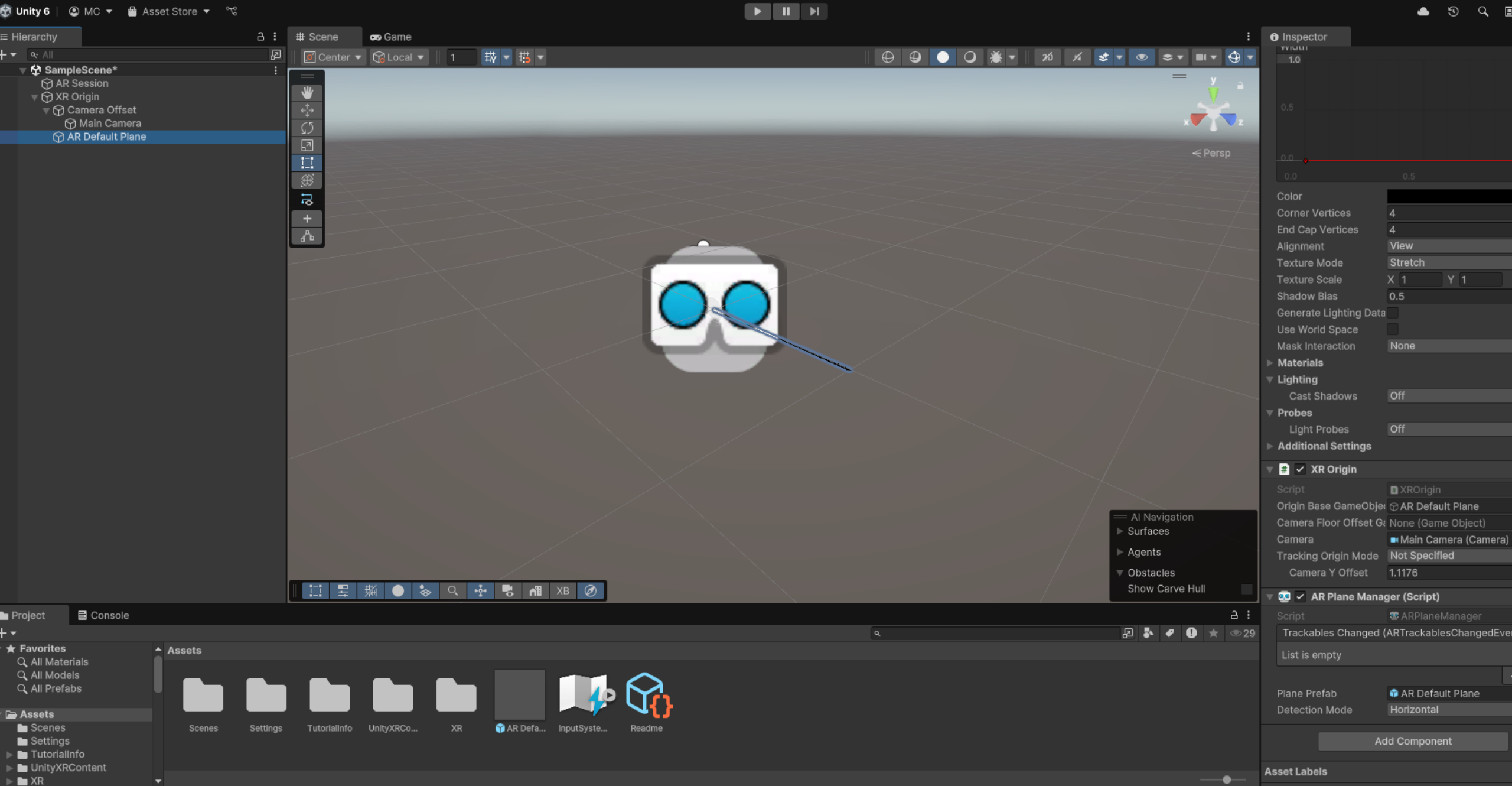
Create AR Session and XR Origin

After adding both an AR Session and XR Origin to the scene, the Hierarchy window will look like the one below, with GameObjects named "AR Session" and "XR Origin". XR Origin has a child GameObject named "Camera Offset", and Camera Offset has a child GameObject named "Main Camera".



Basic Setup for AR Foundation (+AR Core)

Detect Horizontal Planes



Hierarchy

- SampleScene*
- AR Session
 - XR Origin
 - Camera Offset
 - Main Camera
 - AR Default Plane

Inspector

1.0

0.5

0.0

0.0 0.5

Color

Corner Vertices 4

End Cap Vertices 4

Alignment View

Texture Mode Stretch

Texture Scale X 1 Y 1

Shadow Bias 0.5

Generate Lighting Data

Use World Space

Mask Interaction None

Materials

Lighting

Cast Shadows Off

Probes

Light Probes Off

Additional Settings

XR Origin

Script XROrigin

Origin Base GameObject AR Default Plane

Camera Floor Offset GameObject None (Game Object)

Camera Main Camera (Camera)

Tracking Origin Mode Not Specified

Camera Y Offset 1.1176

AR Plane Manager (Script)

Script ARPlaneManager

Trackables Changed (ARTrackablesChangedEvent)

List is empty

Plane Prefab AR Default Plane

Detection Mode Horizontal

Add Component

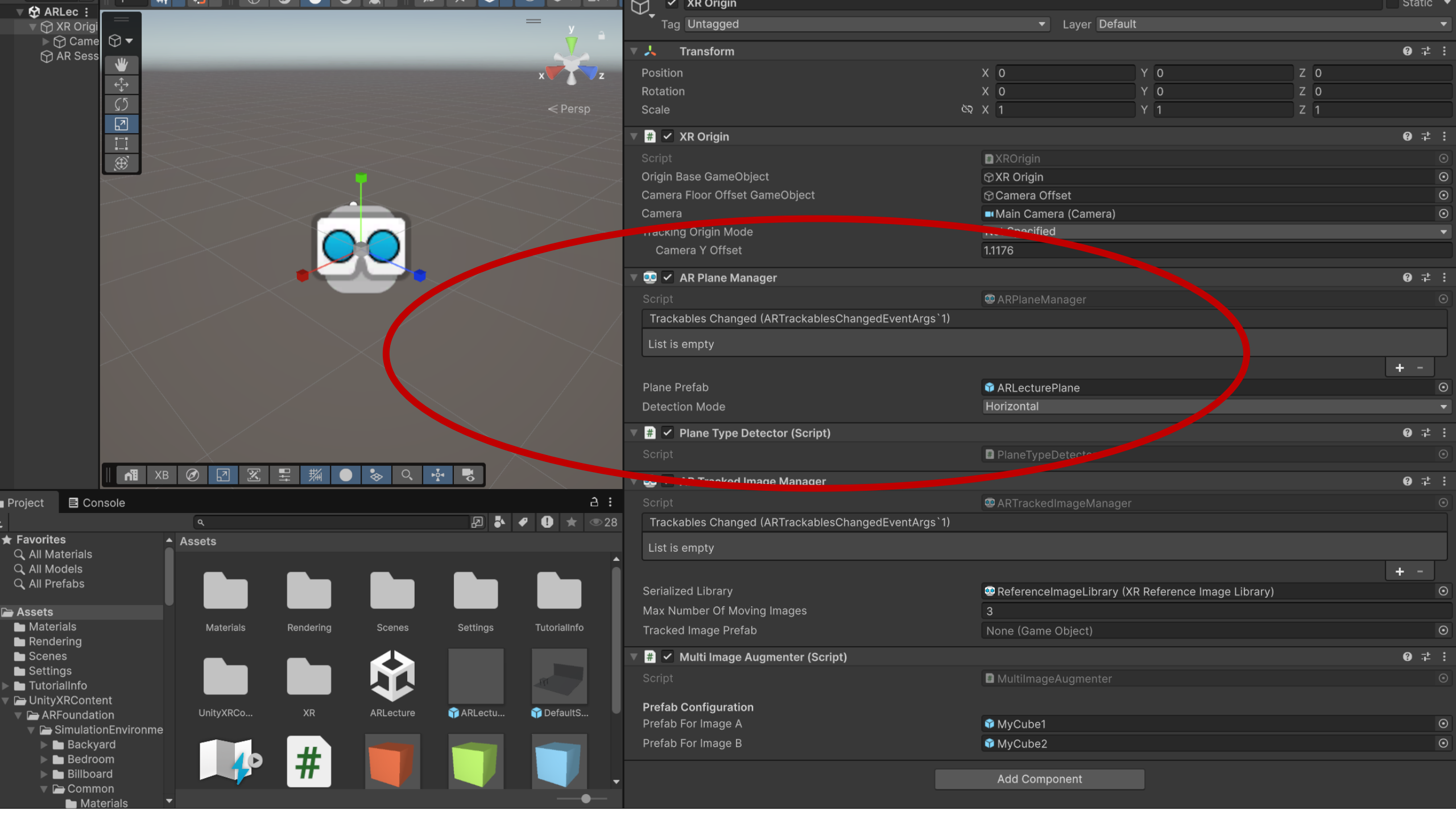
Asset Labels

No Camera Floor Offset GameObject specified for XR Origin, using attached GameObject.

Basic Setup for AR Foundation (+AR Core)

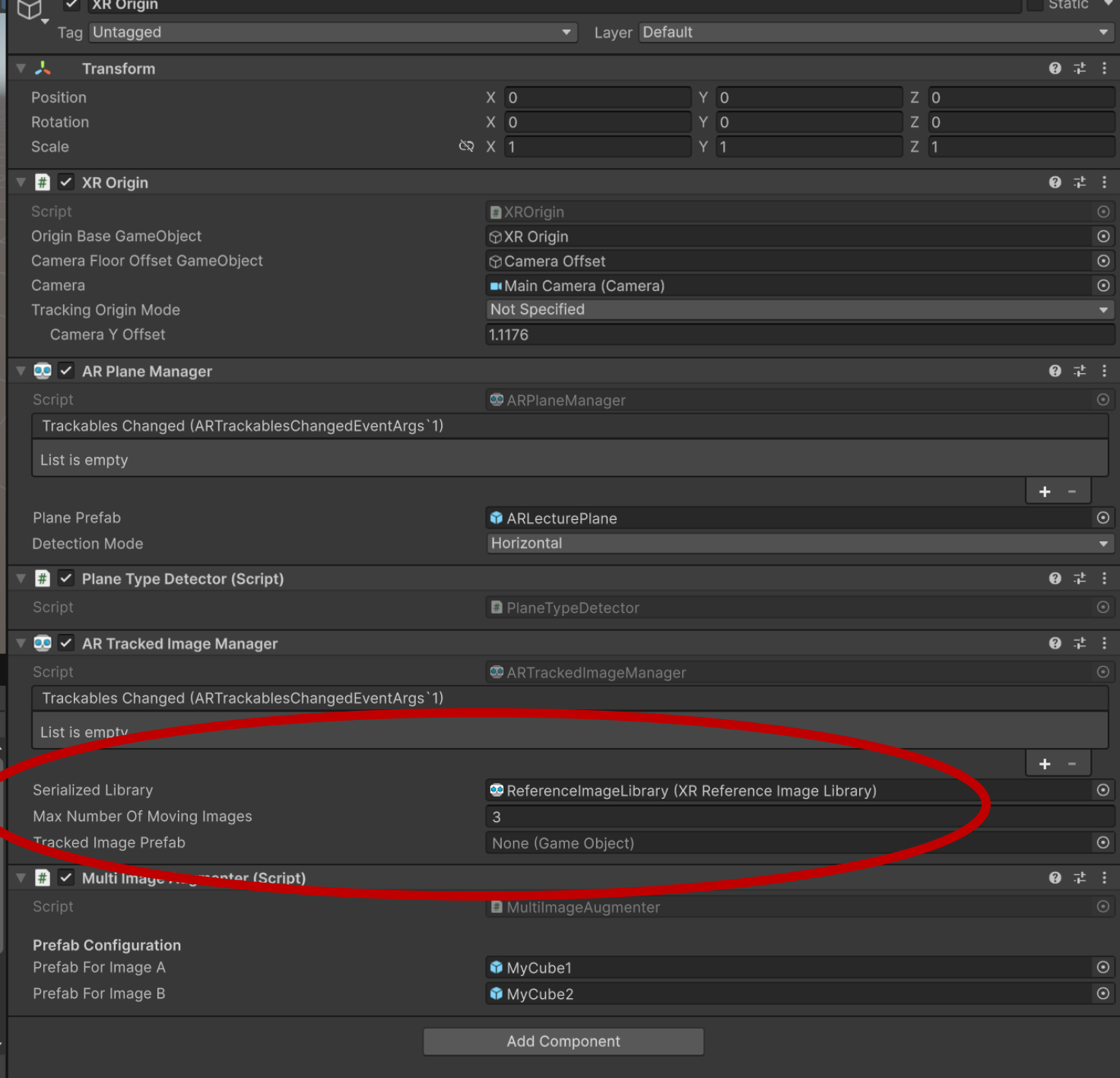
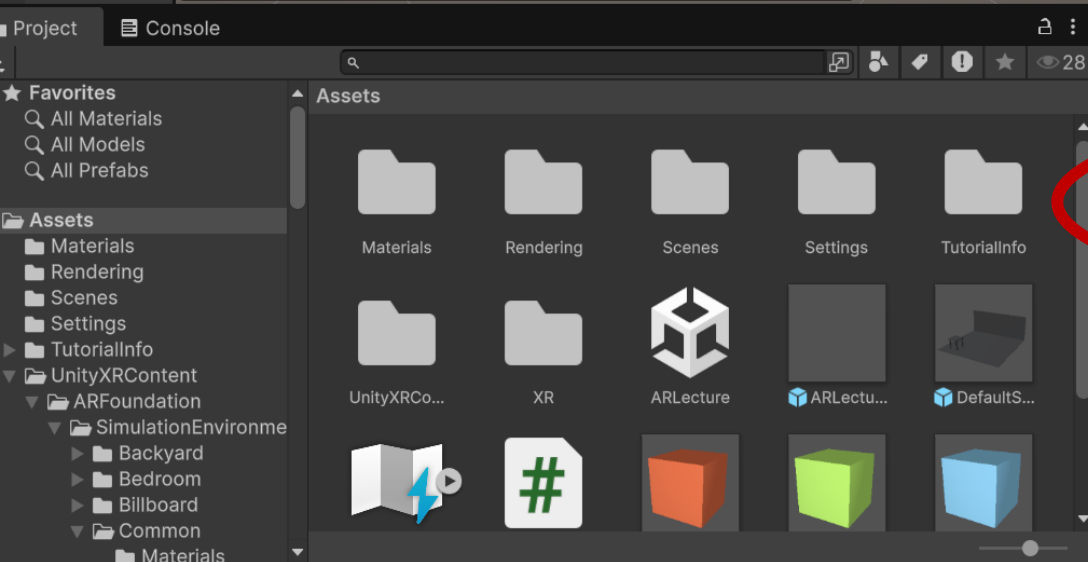
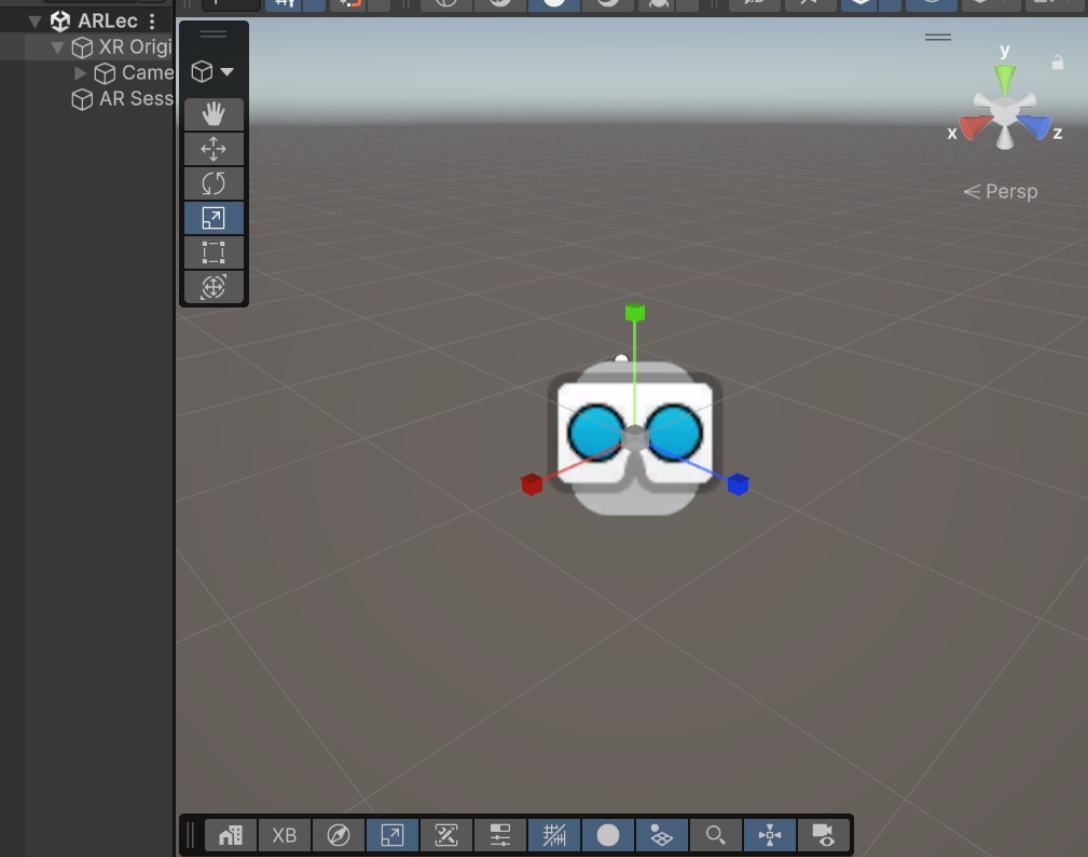
PlaneTypeDetector

Attach a script that shows plane of different orientation in different colors



Basic Setup for AR Foundation (+AR Core)

ImageTargets



Basic Setup for AR Foundation (+AR Core)

Library of Image Targets

Inspector

Reference Image Library (XR Reference Image Library)



Select

Name

unitylogowhiteonblack

Specify Size

☒

Texture Size (pixels)

X 512 Y 512

Physical Size (meters)

X 0.3 Y 0.3

Keep Texture at Runtime

☐



Select

Name

ImageA

Specify Size

☒

Texture Size (pixels)

X 2880 Y 2160

Physical Size (meters)

X 0.3 Y 0.225

Keep Texture at Runtime

☐



Select

Name

ImageB

Specify Size

☒

Texture Size (pixels)

X 1024 Y 681

Physical Size (meters)

X 0.3 Y 0.1995117

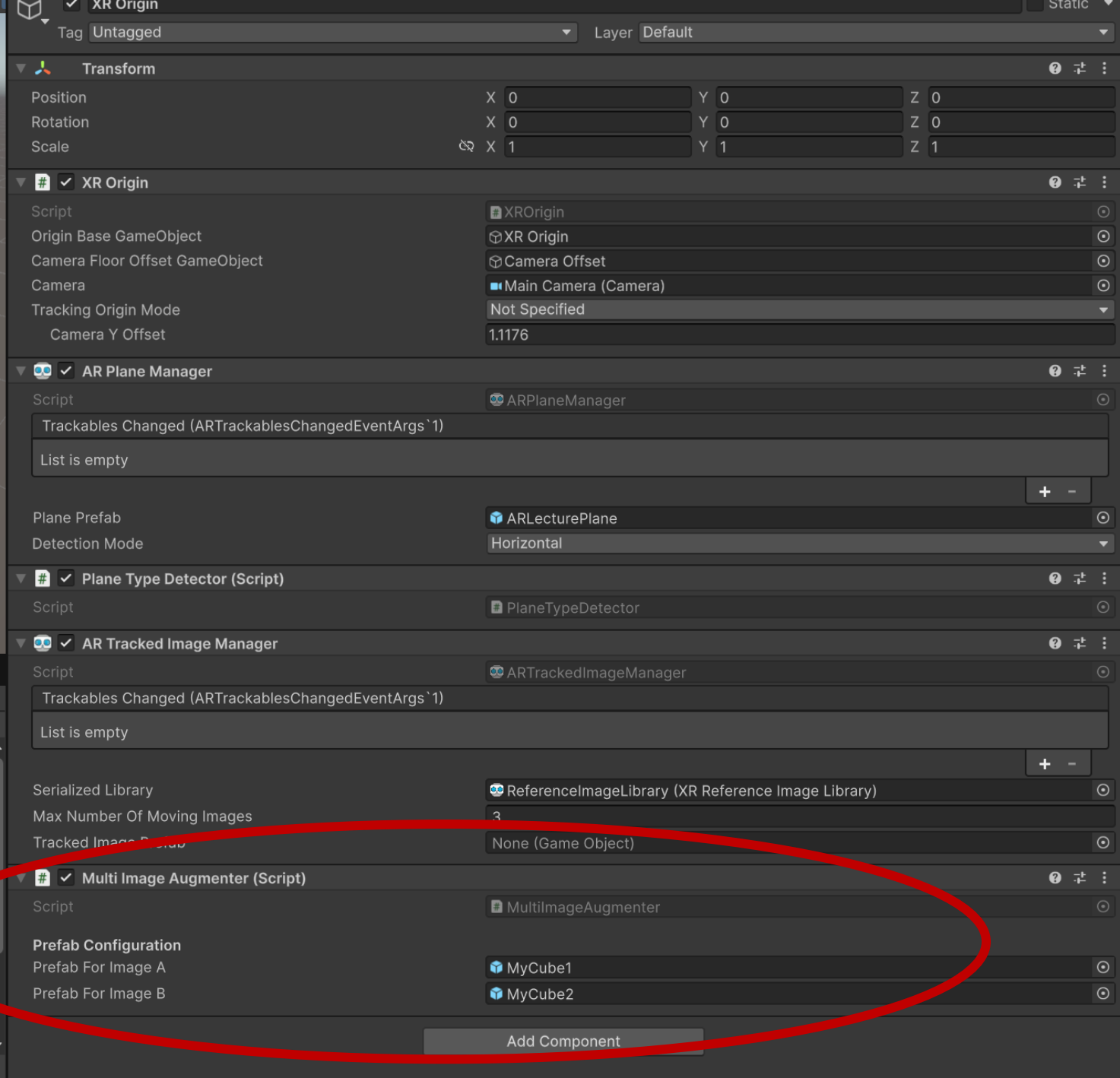
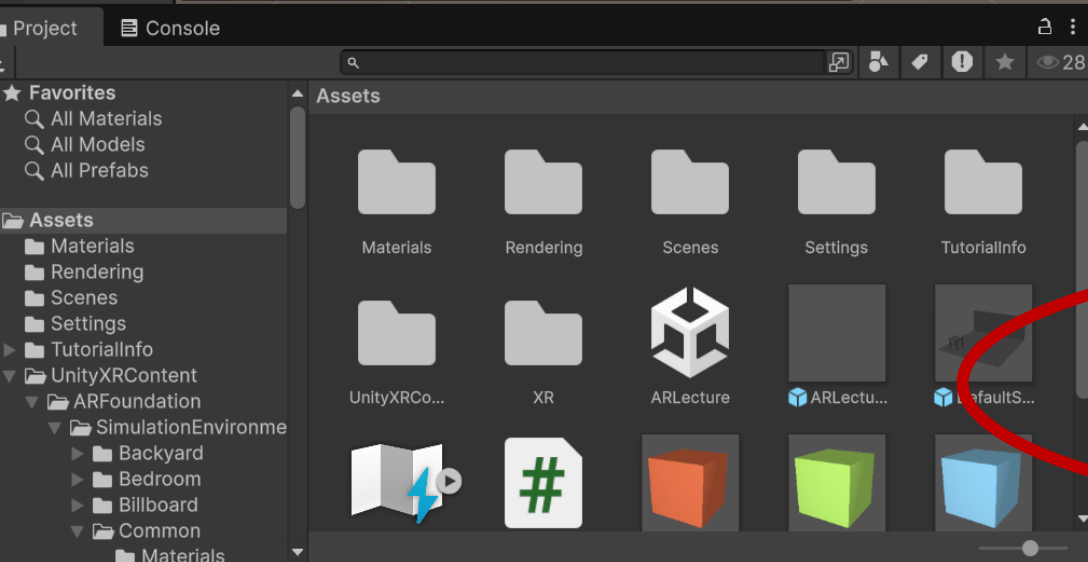
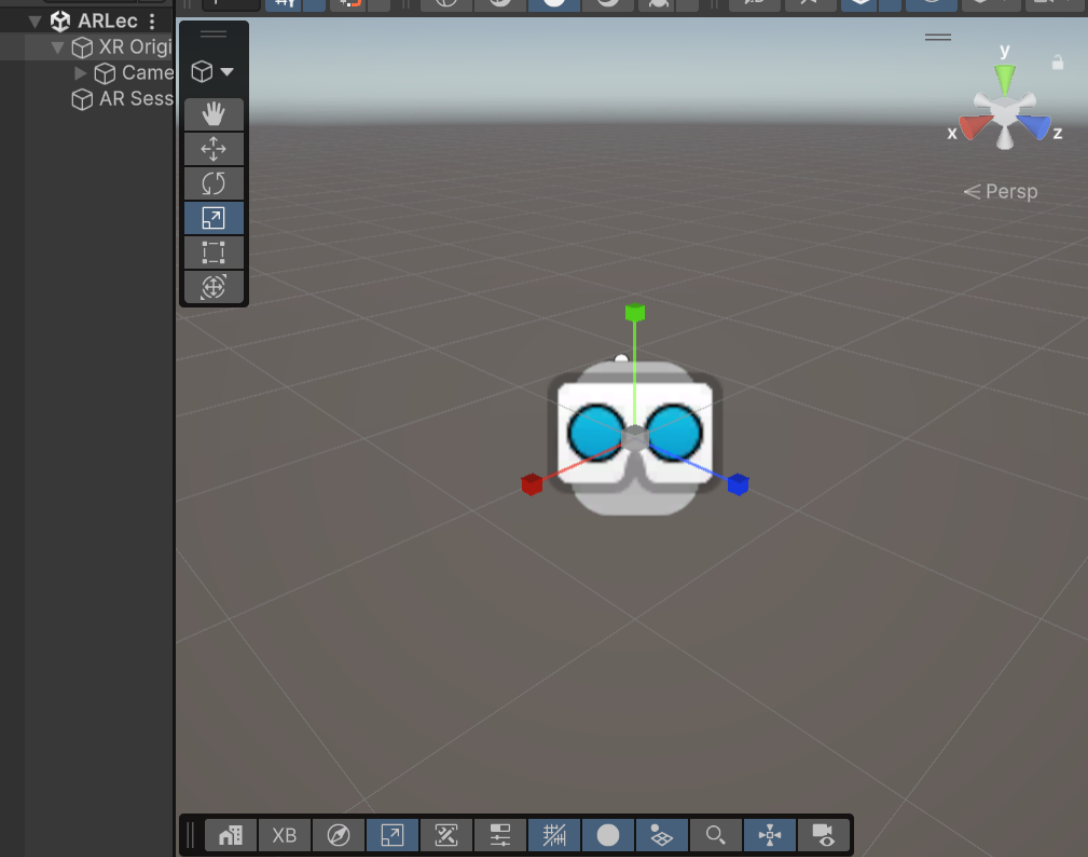
Keep Texture at Runtime

☐

Add Image

Basic Setup for AR Foundation (+AR Core)

ImageTargets - MultiPrefab





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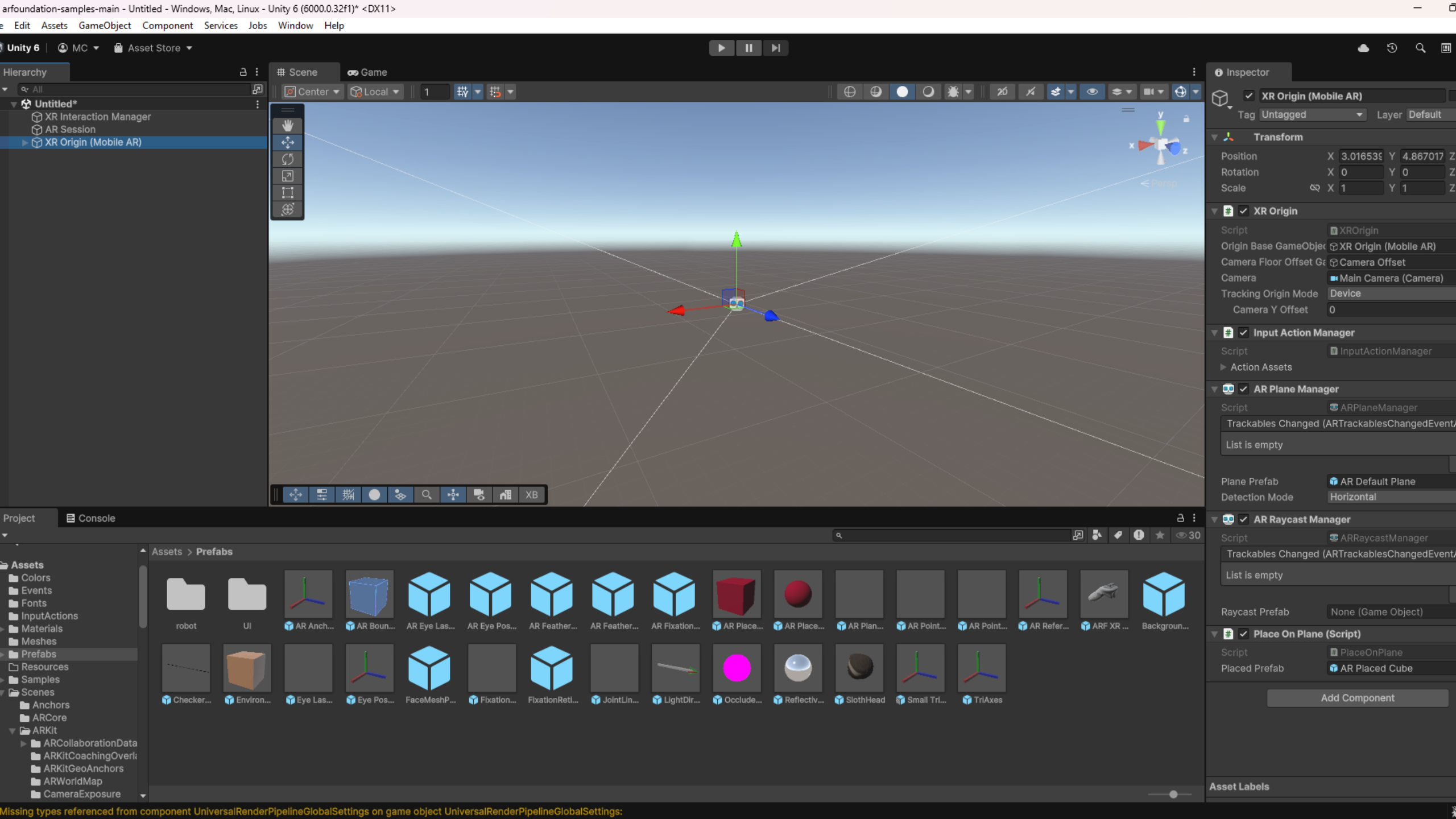
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DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

More advanced features for AR Foundation (+AR Core)

Create a game with AR Core

<https://github.com/Unity-Technologies/arfoundation-samples>

You will have several examples and script

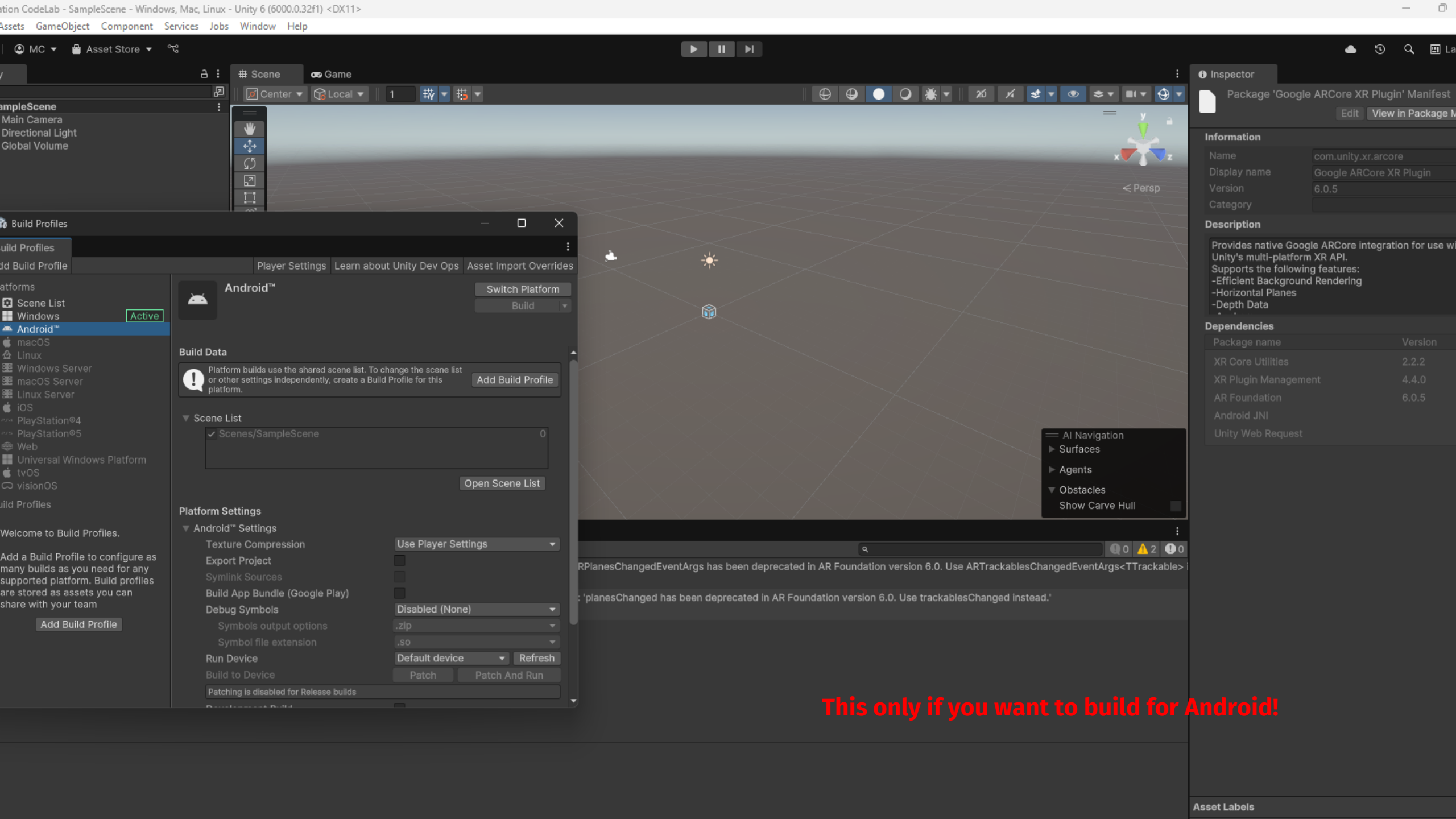


Create a game with AR Core

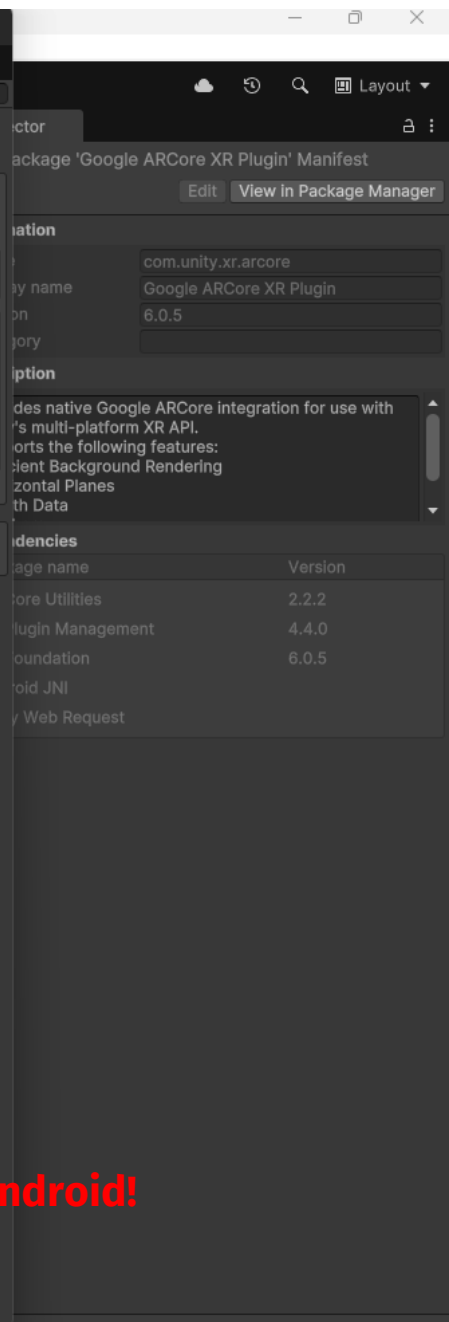
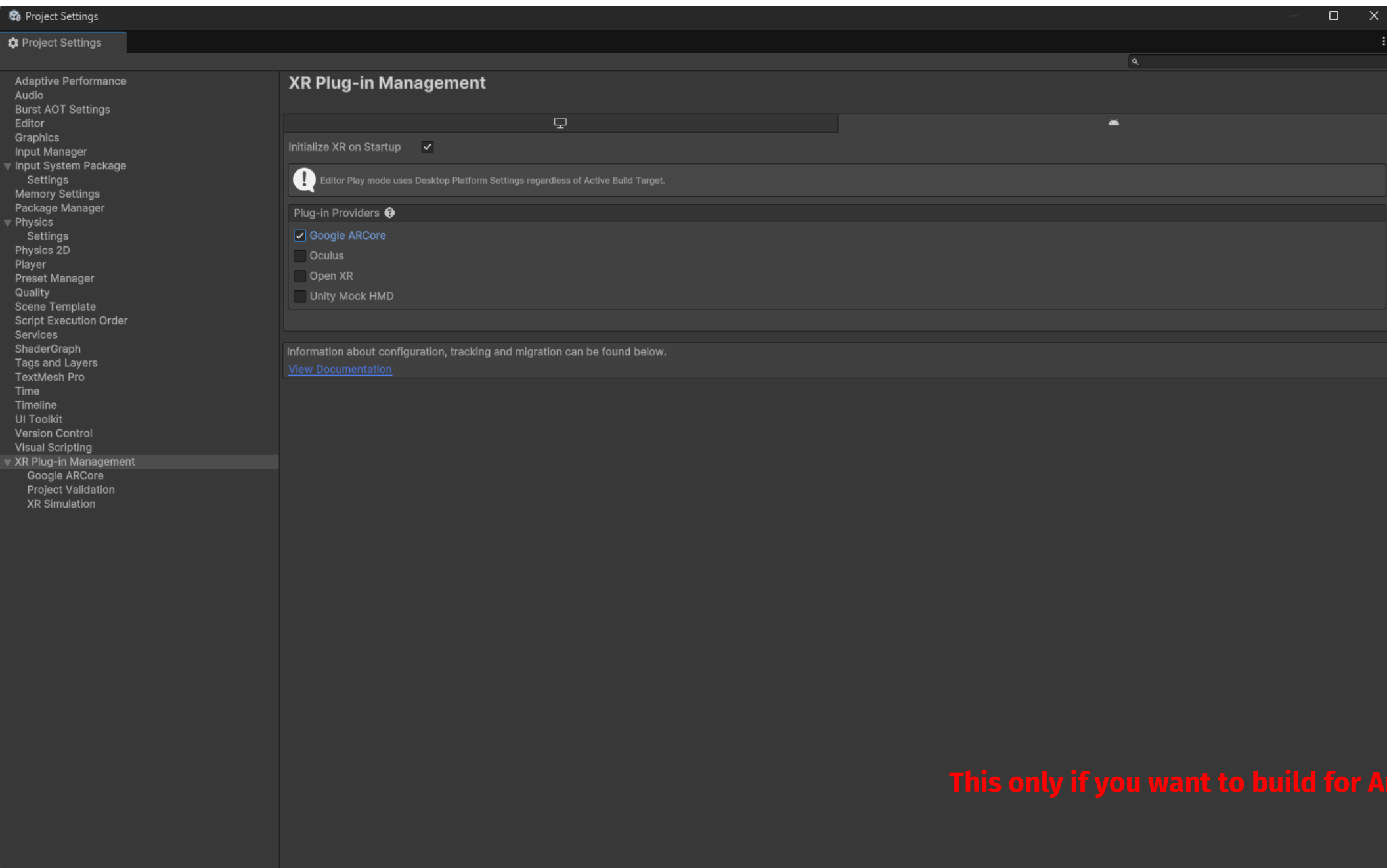
<https://codelabs.developers.google.com/arcore-unity-ar-foundation>

Create a game with AR Core (XRSimulator)

1. New 3D URP project
2. AR Foundation
3. AR Core (Google ARCore XR Plugin)
4. Import codelab-starter-package
5. Enable XR Simulator



This only if you want to build for Android!



This only if you want to build for Android!

Project Settings

Project Settings

Adaptive Performance

Audio

Burst AOT Settings

Editor

Graphics

Input Manager

Input System Package Settings

Memory Settings

Package Manager

Physics Settings

Physics 2D

Player

Preset Manager

Quality

Scene Template

Script Execution Order

Services

ShaderGraph

Tags and Layers

TextMesh Pro

Time

Timeline

UI Toolkit

Version Control

Visual Scripting

XR Plug-in Management

Google ARCore

Project Validation

XR Simulation

Player

Company Name

Product Name

Version

Default Icon

Default Cursor

Cursor Hotspot

X 0

Y 0

Settings for Android

Icon

Resolution and Presentation

Splash Image

Other Settings

Rendering

Color Space*

MSAA Fallback

Auto Graphics API

Graphics API

Vulkan

OpenGL ES3

Require ES3.1

Require ES3.1+AEP

Require ES3.2

Multithreaded Rendering*

Static Batching

Sprite Batching Threshold

Sprite Batching Max Vertex Count

GPU Skinning*

Graphics Jobs (Experimental)

Texture Compression Formats

ASTC

Normal Map Encoding

DXT5nm-style

Remove selection from the list

Layout

Package 'Google ARCore XR Plugin' Manifest

Edit

View in Package Manager

Package Name

com.unity.xr.arcore

Package Name

Google ARCore XR Plugin

Version

6.0.5

Category

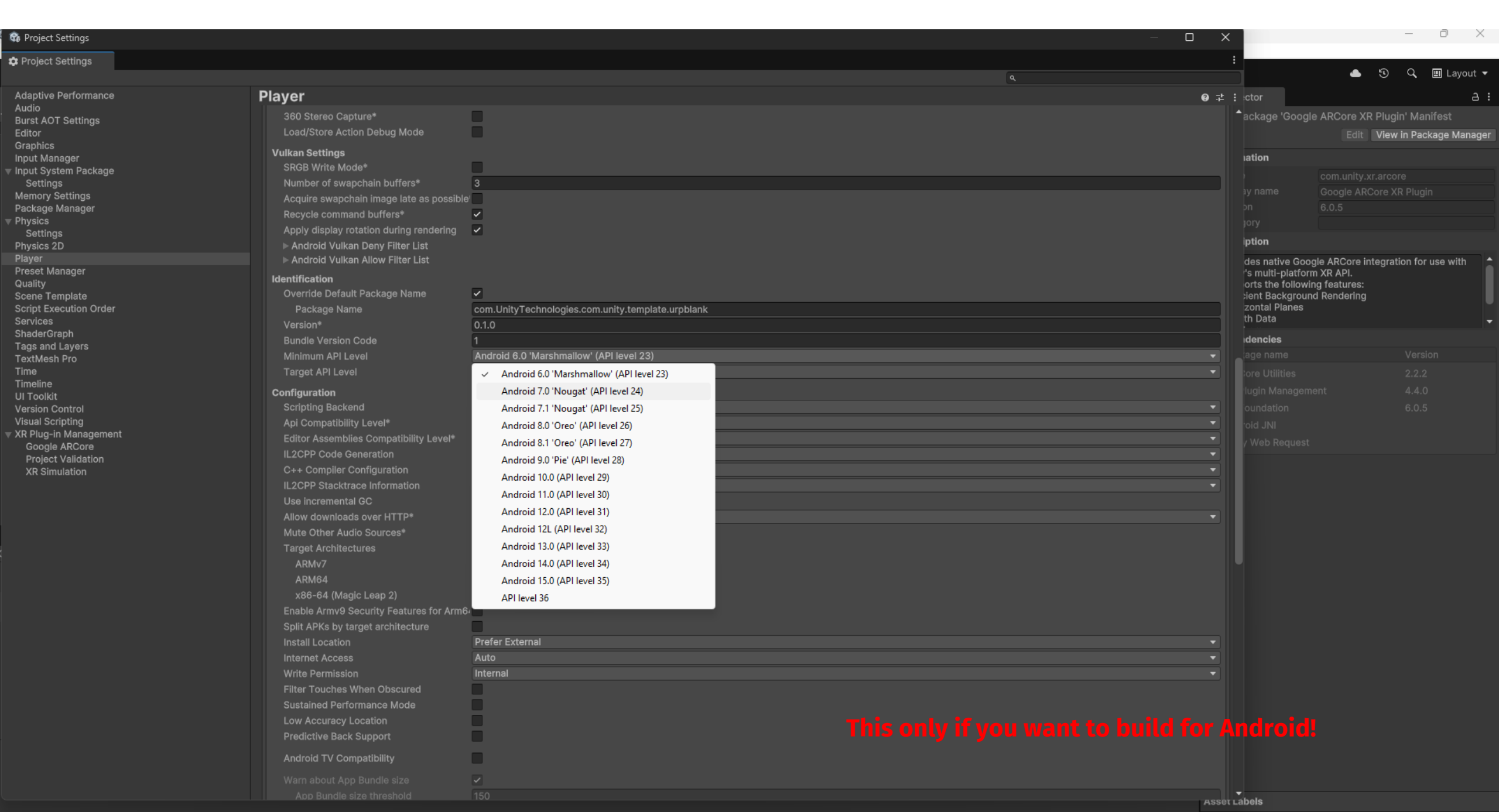
Description

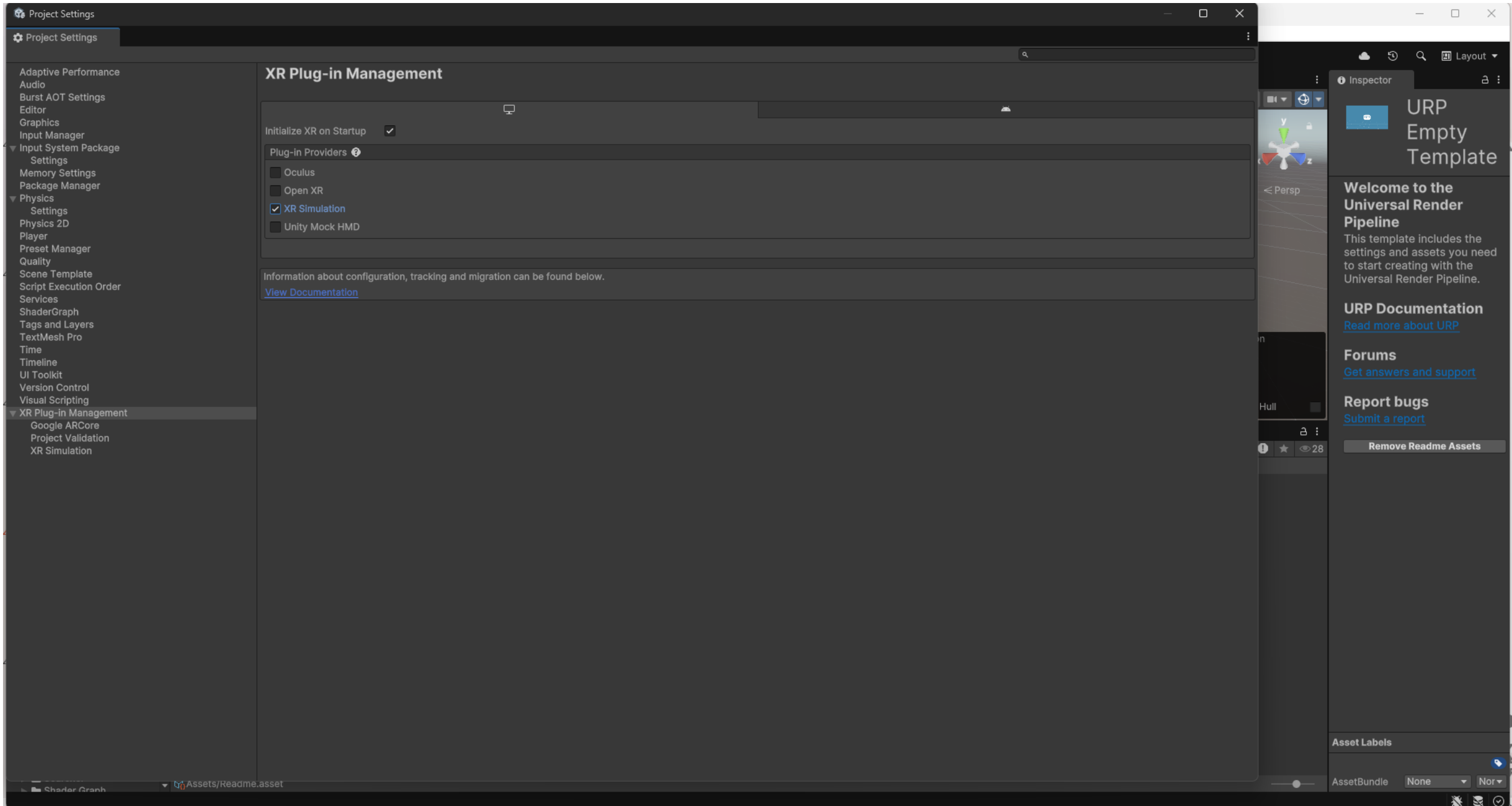
des native Google ARCore integration for use with
's multi-platform XR API.
ports the following features:
Client Background Rendering
Horizontal Planes
Depth Data

Dependencies

Package Name	Version
Core Utilities	2.2.2
Plugin Management	4.4.0
Foundation	6.0.5
Android JNI	
Web Request	

This only if you want to build for Android!





Hierarchy

- SampleScene*
 - Directional Light
 - AR Session
 - XR Origin
 - Camera Offset
 - Main Camera

Scene

Game

Center Local 1

Center Local 1

Center Local 1

Center Local 1

Center Local 1

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Project

Console

Favorites

- All Materials
- All Models
- All Prefabs

Assets

- Scenes
- Settings
- Starter Package
- TutorialInfo
- XR

Packages

- AI Navigation
- AR Foundation
- Burst
- Collections
- Core RP Library
- Custom NUnit
- Editor Coroutines
- Google ARCore XR Plugin

Assets > Settings



DefaultVolu...



Mobile_Ren...



Mobile_RPA...



PC_Renderer



PC_RPAsset



SampleSce...



UniversalR...

Assets/Settings/Mobile_Renderer.asset



Inspector

Mobile_Renderer (Universal Render Data)

Open

Filtering

Opaque Layer Mask Everything

Transparent Layer M Everything

Rendering

Rendering Path Forward+

Depth Priming Mo Disabled

Depth Texture Mode After Opaques

Depth Attachment F Default

Depth Texture Form Default

Shadows

Transparent Receive

Post-processing

Enabled

Data PostProcessData (Post Process D

Overrides

Stencil

Compatibility

Intermediate Texture Auto

Renderer Features

AR Background Renderer Feature (AR Backgrou

Add Renderer Feature

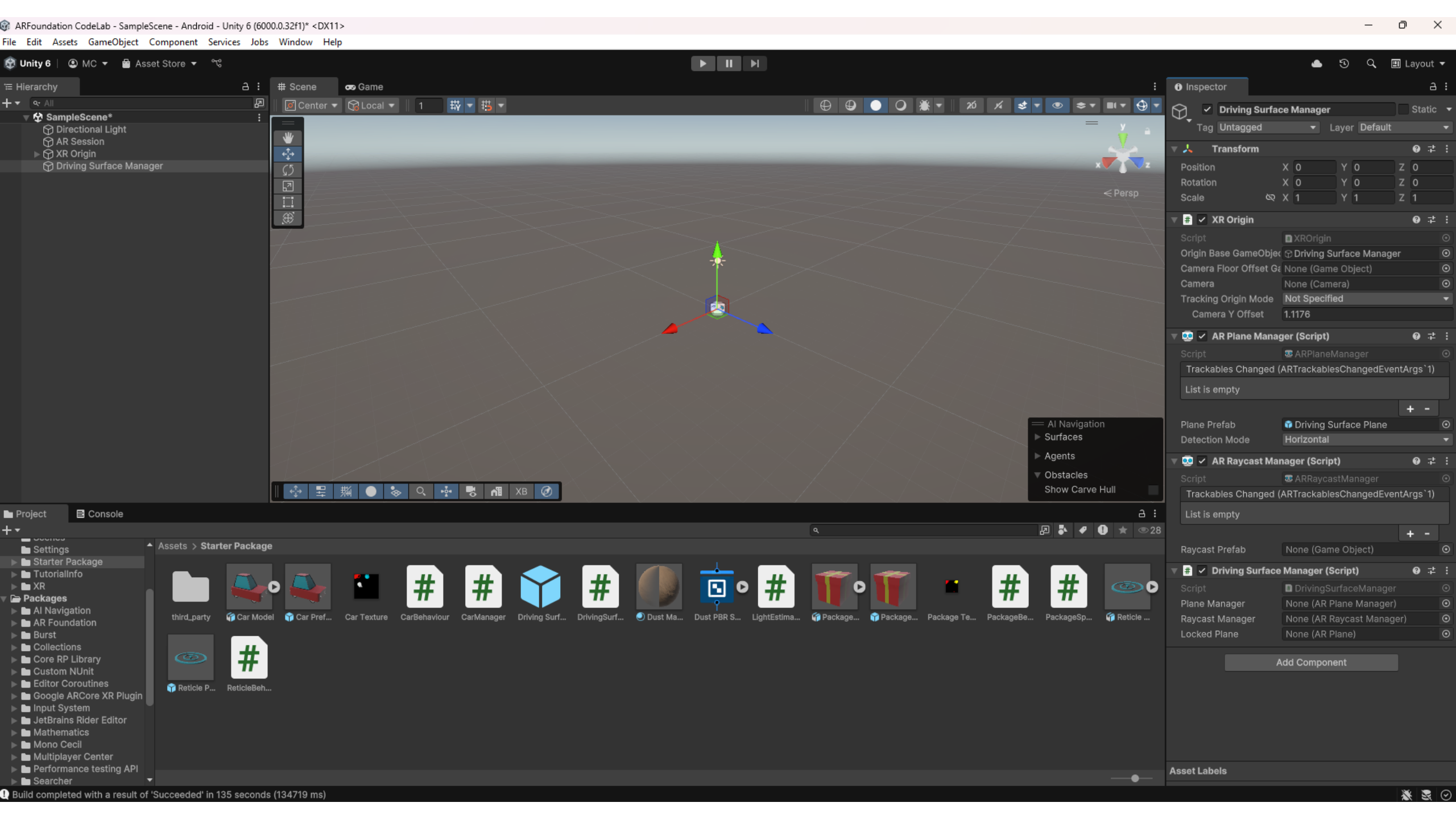
Asset Labels

AssetBundle None

None

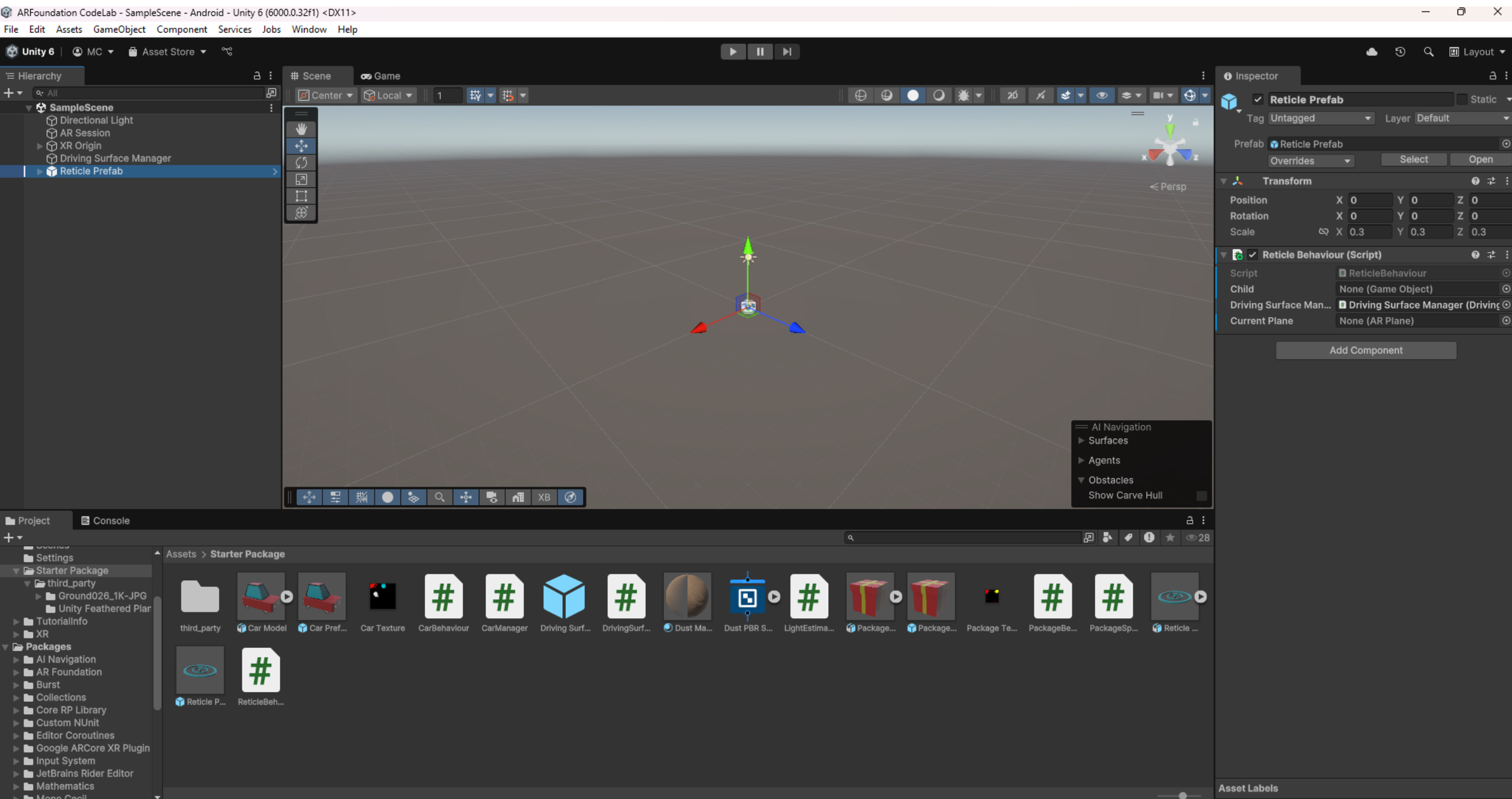
Create a game with AR Core

Detect and visualize the plane



Create a game with AR Core

Add a Reticle that follows the camera



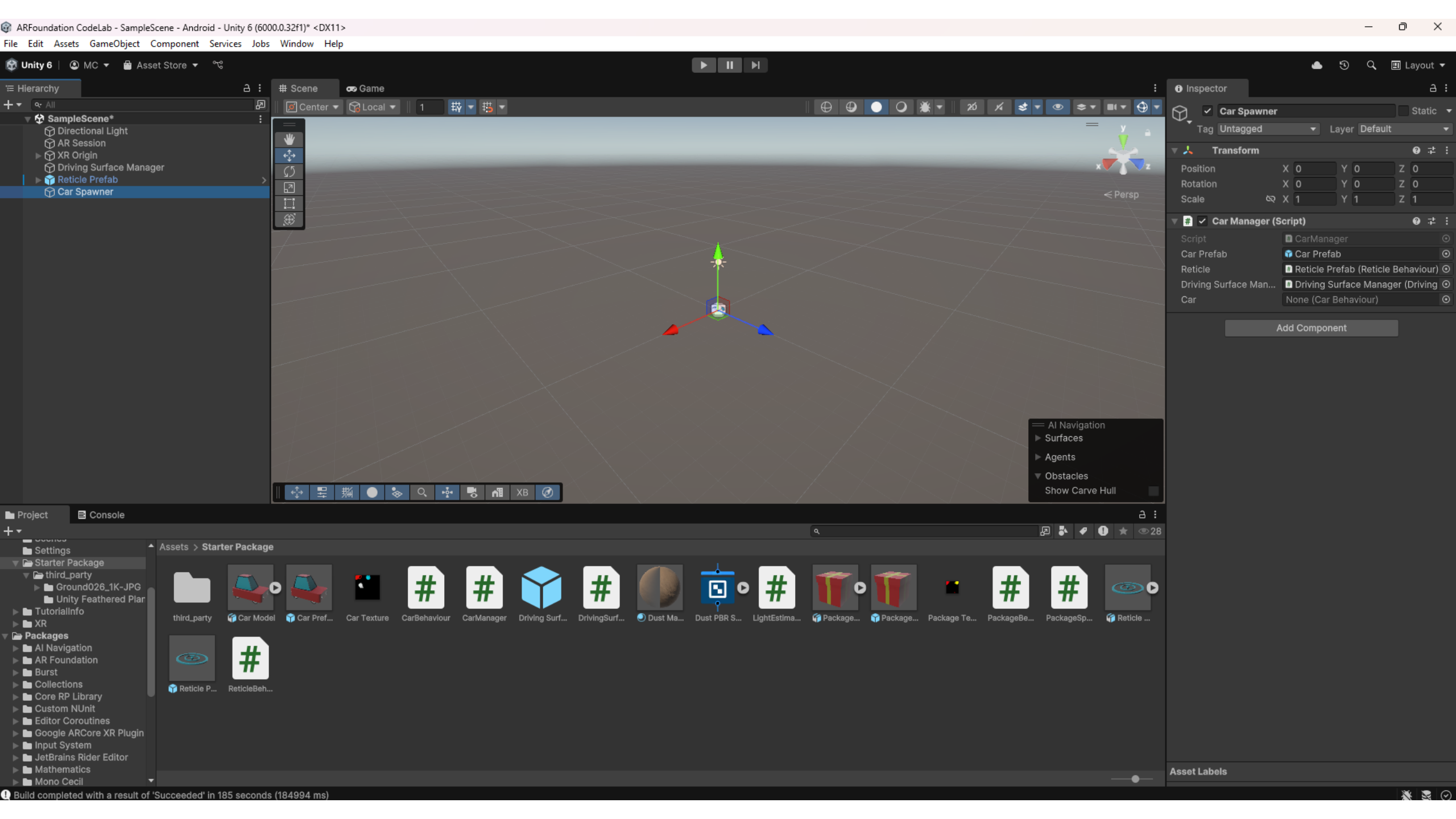
```
26 public class ReticleBehaviour : MonoBehaviour
```

```
0 references
```

```
39 private void Update()
40 {
41     // TODO: Conduct a ray cast to position this object.
42     var screenCenter = Camera.main.ViewportToScreenPoint(new Vector3(0.5f, 0.5f));
43     var hits = new List<ARRaycastHit>();
44     DrivingSurfaceManager.RaycastManager.Raycast(screenCenter, hits, TrackableType.PlaneWithinBounds);
45
46     CurrentPlane = null;
47     ARRaycastHit? hit = null;
48     if (hits.Count > 0)
49     {
50         // If you don't have a locked plane already...
51         var lockedPlane = DrivingSurfaceManager.LockedPlane;
52         hit = lockedPlane == null
53             ? hits[0]
54             // Otherwise use the locked plane, if it's there.
55             : hits.SingleOrDefault(x => x.trackableId == lockedPlane.trackableId);
56     }
57
58     if (hit.HasValue)
59     {
60         CurrentPlane = DrivingSurfaceManager.PlaneManager.GetPlane(hit.Value.trackableId);
61         // Move this reticle to the location of the hit.
62         transform.position = hit.Value.pose.position;
63     }
64     Child.SetActive(CurrentPlane != null);
65 }
66
67 }
68 }
```

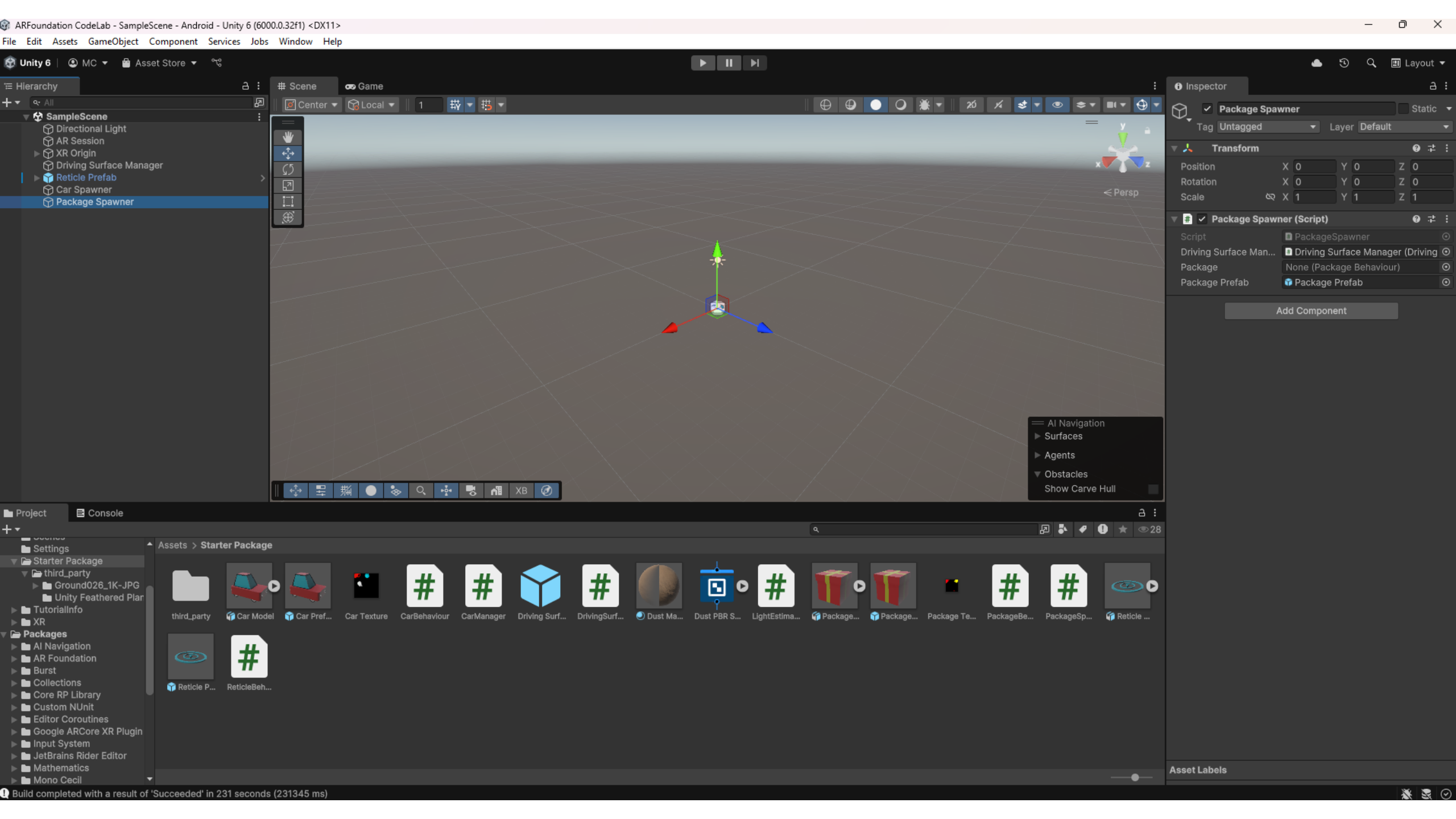
Create a game with AR Core

Add a car



Create a game with AR Core

Spawn targets (package)



Create a game with AR Core

Visual coherence: light estimation

The screenshot displays the Unity 6 development environment. The central 3D Viewport shows a scene with a grid floor and a camera. The Hierarchy panel on the left lists objects under 'SampleScene*', including 'Directional Light', 'AR Session', 'XR Origin', 'Camera Offset', 'Main Camera', 'Driving Surface Manager', 'Reticle Prefab', 'Car Spawner', and 'Package Spawner'. The Inspector panel on the right shows settings for the 'AR Camera Manager (Script)', including 'Script' (ARCameraManager), 'Auto Focus' (checked), 'Image Stabilization' (unchecked), 'Light Estimation' (Everything), 'Facing Direction' (World), and 'Render Mode' (Any). The Project panel at the bottom shows the 'Assets > Starter Package' folder containing various assets like 'third_party', 'Car Model', 'Car Prefab', 'Car Texture', 'CarBehaviour', 'CarManager', 'Driving Surf...', 'Dust Ma...', 'Dust PBR S...', 'LightEstima...', 'Package...', 'PackageTe...', 'PackageBe...', 'PackageSp...', 'Reticle P...', and 'ReticleBeh...'. A context menu is visible in the 3D Viewport, showing options like 'AI Navigation', 'Surfaces', 'Agents', 'Obstacles', and 'Show Carve Hull'.

Inspector

Field of View

Clipping Planes

Near 0.1

Far 20

Physical Camera

Rendering

Renderer Default Renderer (PC_Renderer)

Post Processing

Anti-aliasing No Anti-aliasing

Stop NaNs

Dithering

Render Shadows

Priority 0

Opaque Texture Use settings from Render Pipeline

Depth Texture Use settings from Render Pipeline

Culling Mask Everything

Occlusion Culling

Stack

Environment

Background Type Solid Color

Background

Volumes

Update Mode Use Pipeline Settings

Volume Mask Default

Volume Trigger None (Transform)

Output

Audio Listener

AR Camera Manager (Script)

Script ARCameraManager

Auto Focus

Image Stabilization

Light Estimation Everything

Facing Direction World

Render Mode Any

AR Camera Background (Script)

Use Custom Material

Tracked Pose Driver (Input System)

Script TrackedPoseDriver

Tracking Type Rotation And Position

Update Type Update And Before Render

Ignore Tracking State

Position Input Action

Asset Labels

Build completed with a result of 'Succeeded' in 231 seconds (231345 ms)

Modify the directional light

1. In **Hierarchy**, select the **Directional Light** object.
2. Add the **LightEstimation** component to it. This component from the Starter Package provides some boilerplate for subscribing to lighting changes.
3. In the **FrameReceived()** function, add:

```
ARLightEstimationData lightEstimation = args.lightEstimation;

if (lightEstimation.averageBrightness.HasValue)
    Light.intensity = lightEstimation.averageBrightness.Value;

if (lightEstimation.averageColorTemperature.HasValue)
    Light.colorTemperature = lightEstimation.averageColorTemperature.Value;

if (lightEstimation.colorCorrection.HasValue)
    Light.color = lightEstimation.colorCorrection.Value;

if (lightEstimation.mainLightDirection.HasValue)
    Light.transform.rotation = Quaternion.LookRotation(lightEstimation.mainLightDirection);

if (lightEstimation.mainLightColor.HasValue)
    Light.color = lightEstimation.mainLightColor.Value;

if (lightEstimation.mainLightIntensityLumens.HasValue)
    Light.intensity = lightEstimation.averageMainLightBrightness.Value;

if (lightEstimation.ambientSphericalHarmonics.HasValue)
{
    RenderSettings.ambientMode = AmbientMode.Skybox;
    RenderSettings.ambientProbe = lightEstimation.ambientSphericalHarmonics.Value;
}
```

Check out [the AR Foundation documentation on ARLightEstimationData](#) for more details on each parameter.

Create a game with AR Core

Some notes

In the ReticleBehavior.cs script, in Update() modify

if (hits.Length > 0) to if (hits.Count > 0)

In the DrivingSurfaceManager.cs script, in Start() add

```
RaycastManager = GetComponent <ARRaycastManager>();
```

Player

Lightmap Streaming	☒
Streaming Priority	0
Frame Timing Stats	☐
OpenGL: Profiler GPU Recorders	☒

 On OpenGL, Profiler GPU Recorders may disable the GPU Profiler.

Allow HDR Display Output*	☐
Use HDR Display Output*	☐
Swap Chain Bit Depth	Bit Depth 10
Virtual Texturing (Experimental)*	☐
360 Stereo Capture*	☐
Load/Store Action Debug Mode	☐

Vulkan Settings

SRGB Write Mode*	☐
Number of swapchain buffers*	3
Get swapchain image late as possible*	☐
Recycle command buffers*	☒

D3D12 Settings

Device Filtering Asset	None (D3D12 Device Filter Lists)
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Configuration

Scripting Backend	Mono
Api Compatibility Level*	.NET Standard 2.1
Editor Assemblies Compatibility Level*	Default (.NET Framework)
IL2CPP Code Generation	Optimize for runtime speed
C++ Compiler Configuration	Release
IL2CPP Stacktrace Information	Method Name
Use Incremental GC*	☒
Allow downloads over HTTP*	Not allowed
Active Input Handling*	Both

Exercise

Augment your room with flowers

Starting from the previous example, create a simple application that places flowers (possibly different prefabs) on the floor of a room (position after mouse click)

Test it with different XR environments